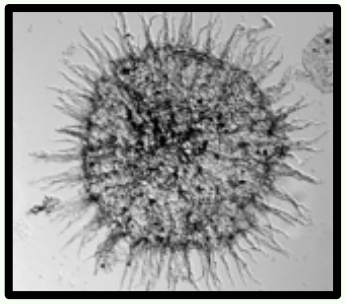
Part 3
Origin of Animals

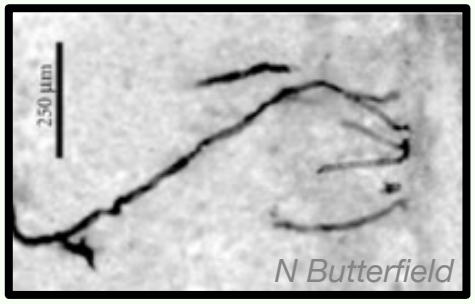
3.1 Domains and Kingdoms of Life

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Photosynthesing microorganisms
Mostly prokaryotes, some eukaryotes
Mostly single cells, some multicellular

Swayer/Smithsonian



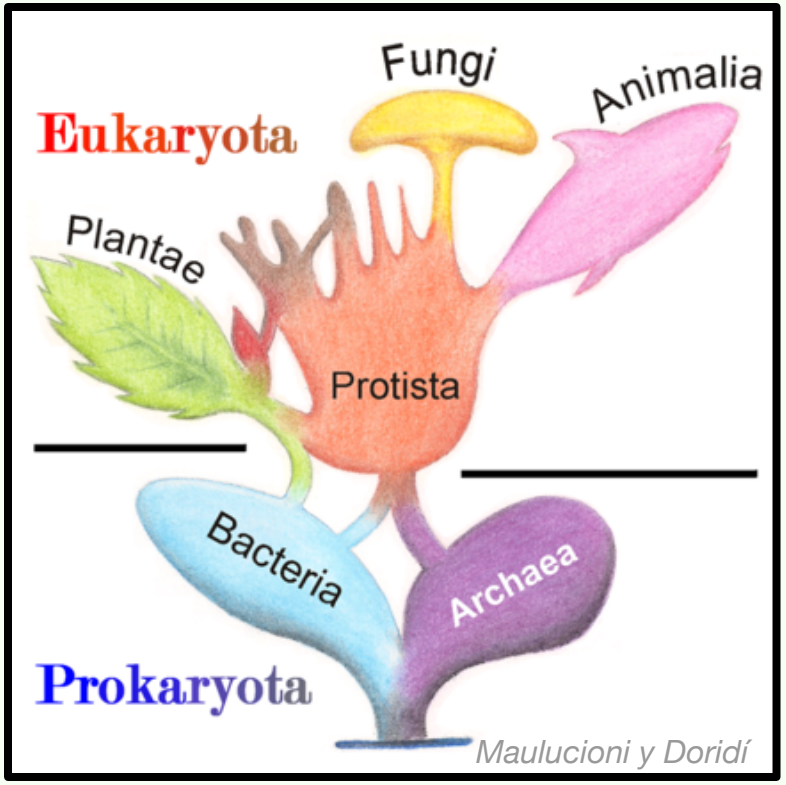


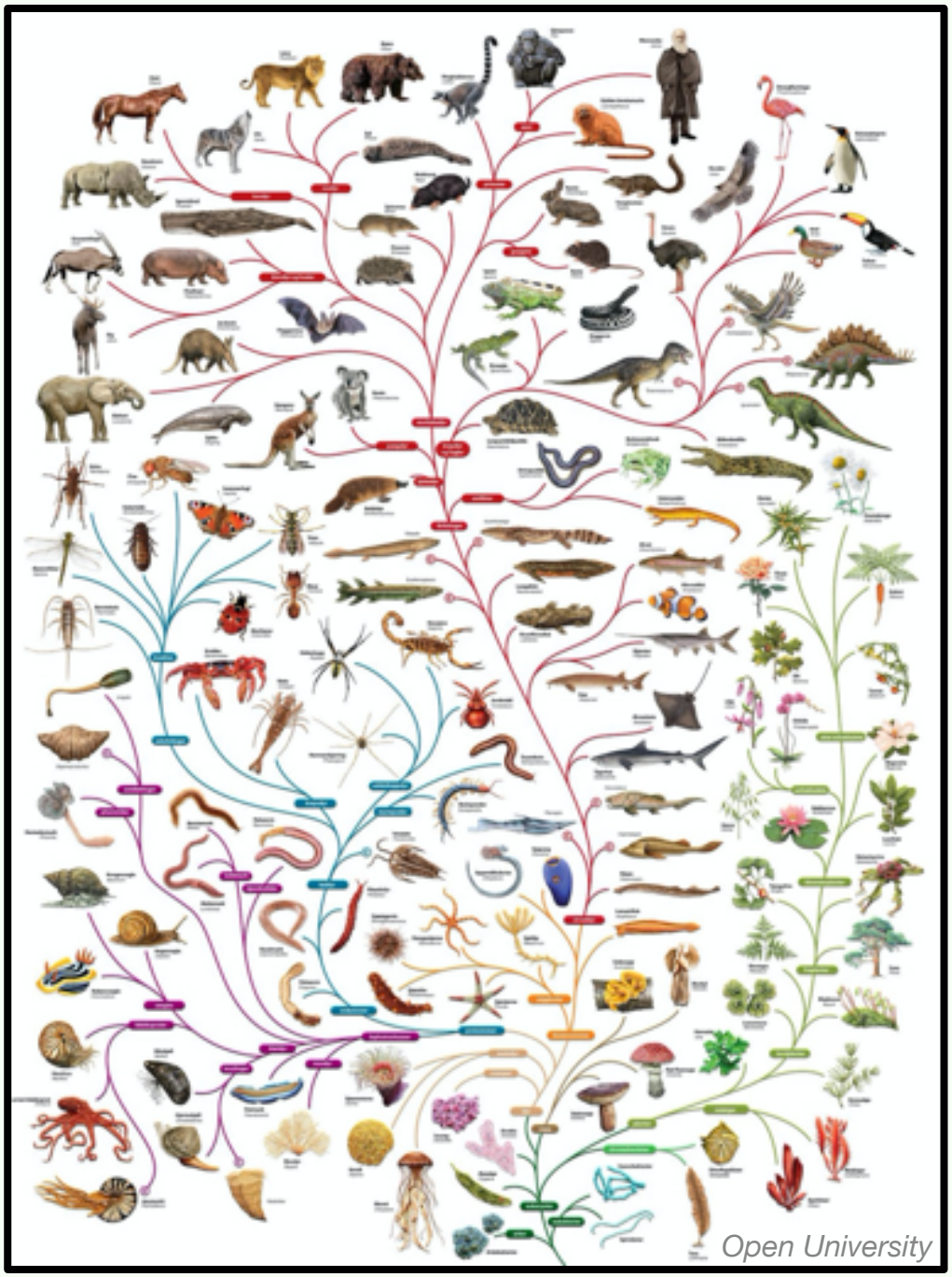
How and when did life make this transition?

Plants

Animals

Fungi



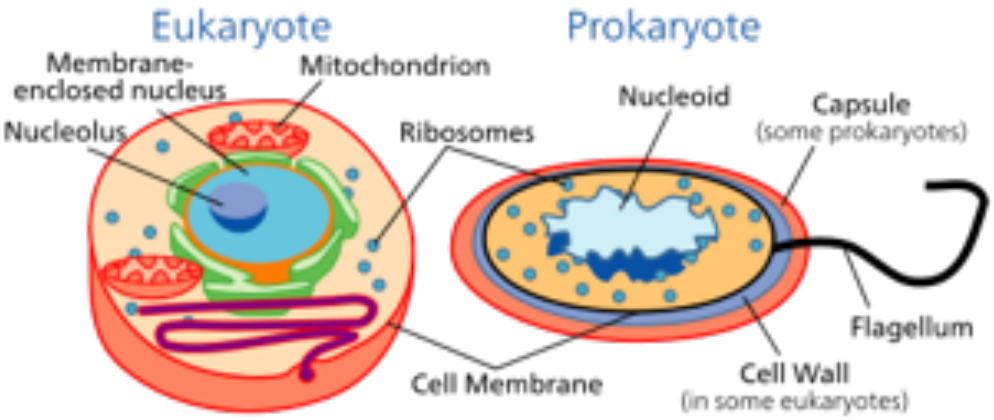


Classification of Life: Domain

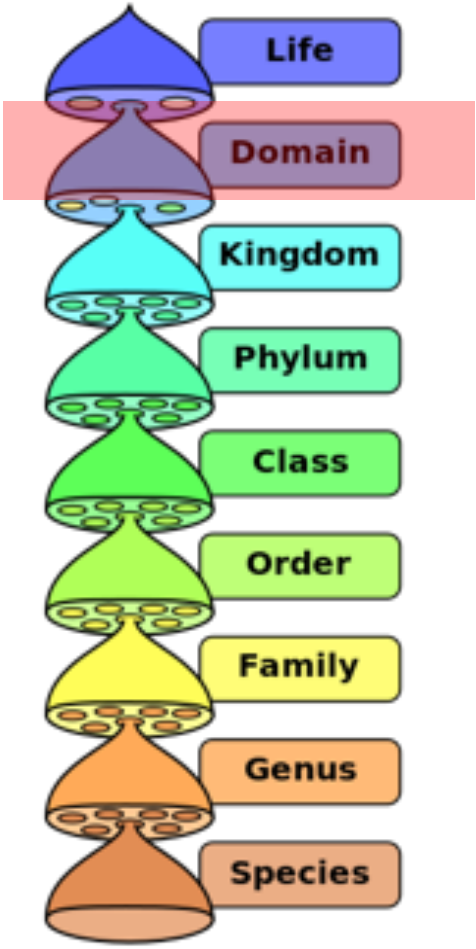
Bacteria:
unicellular prokaryotes with fatty ester lipids

Archaea:
unicellular prokaryotes with isoprene ether lipids

Eukaryotes:
have a cell nucleus and membrane bound organelles, can be multicellular



Taxonomy:
Hierarchical Classification

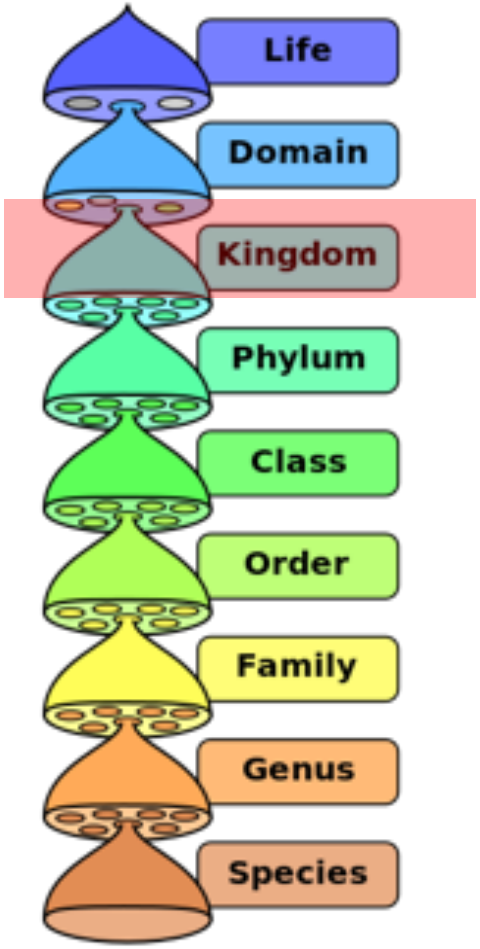


Classification of Life: Kingdom

Plantae:
Multicellular autotrophs with chloroplasts and cellulose cell walls

Fungi:
Multicellular and unicellular heterotrophs with chitin cell walls

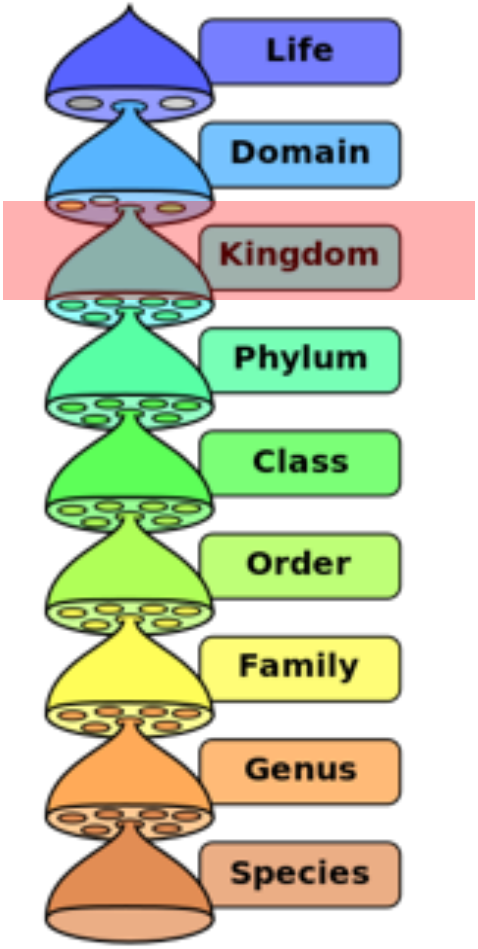
Taxonomy:
Hierarchical Classification



Classification of Life: Kingdom

Animalia (Metazoa): What is an animal?
Multicellular heterotrophs without a cell wall
Motility: they move, at least in part of the life cycle

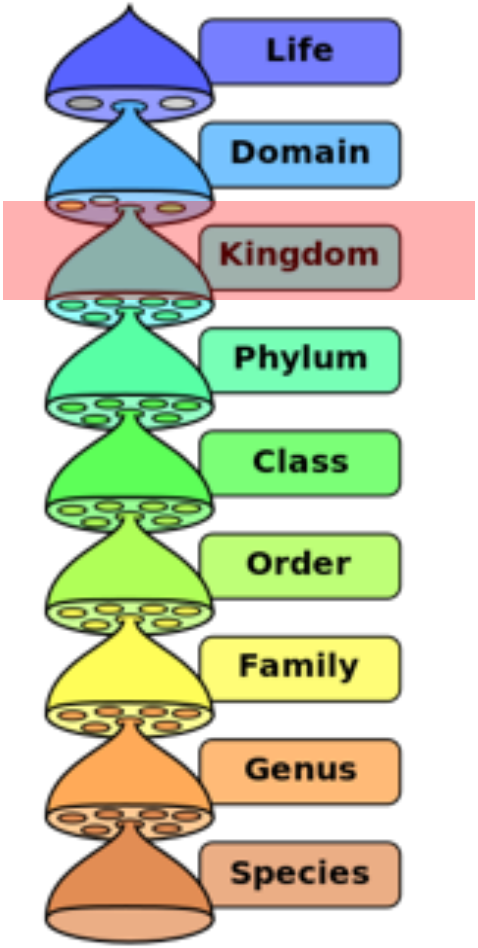
Taxonomy:
Hierarchical Classification



Classification of Life: Kingdom

Animalia (Metazoa): What is an animal?
Most are **bilaterians**: left side, right side & a head with mouth and through gut

Taxonomy:
Hierarchical Classification



Origins of Animals
3.1 Domains of Life

Intended Learning Objective: Define major domains and kingdoms of life (plants, fungi and animals) in the context of hierarchical biological taxonomy

Question: What are some important sub-groups of animals or plants?
5 minutes

Robert Sansom, University of Manchester

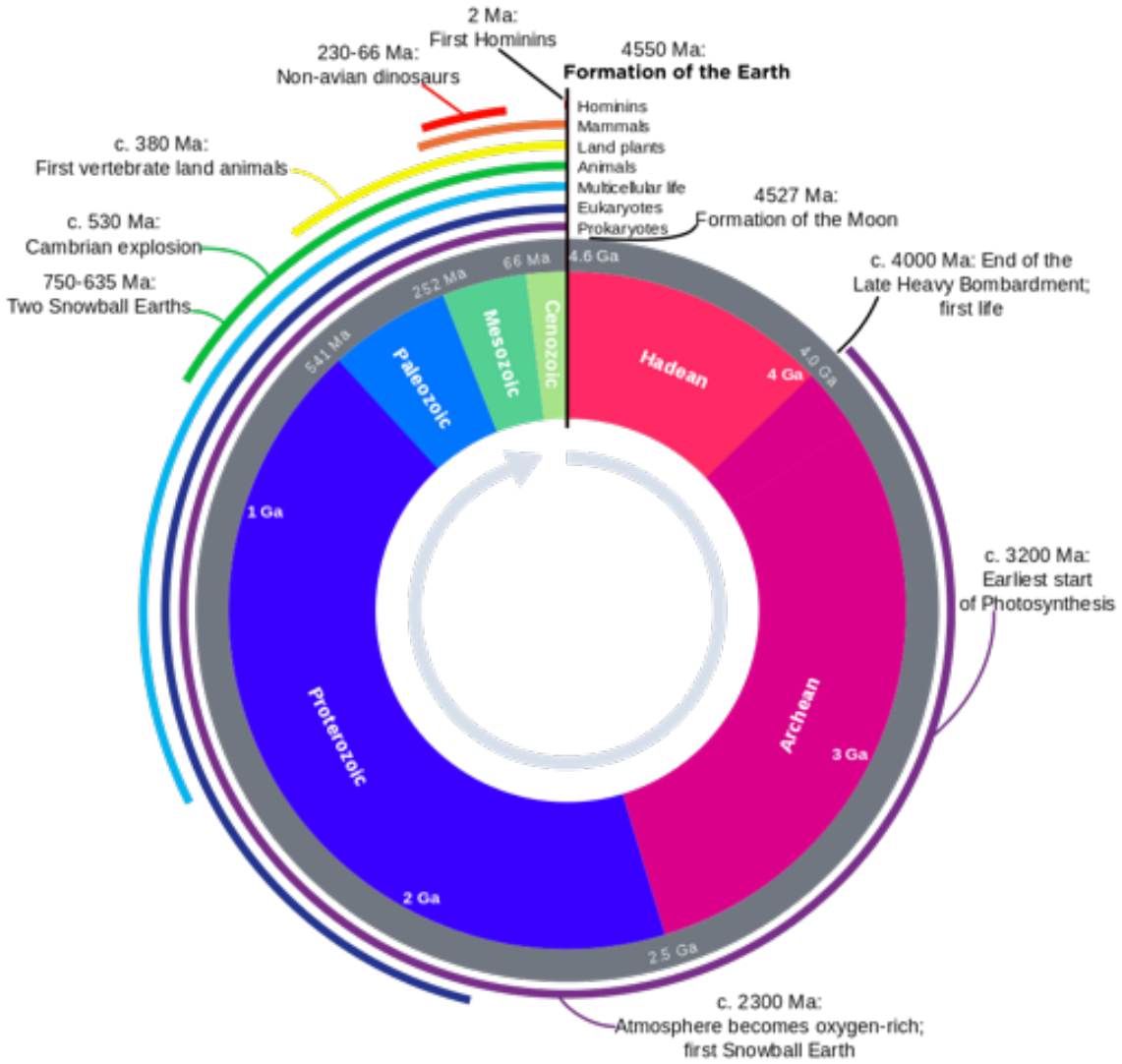
Part 3
Origin of Animals

3.2 Ediacaran Animals?

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

How and when did life make this transition?

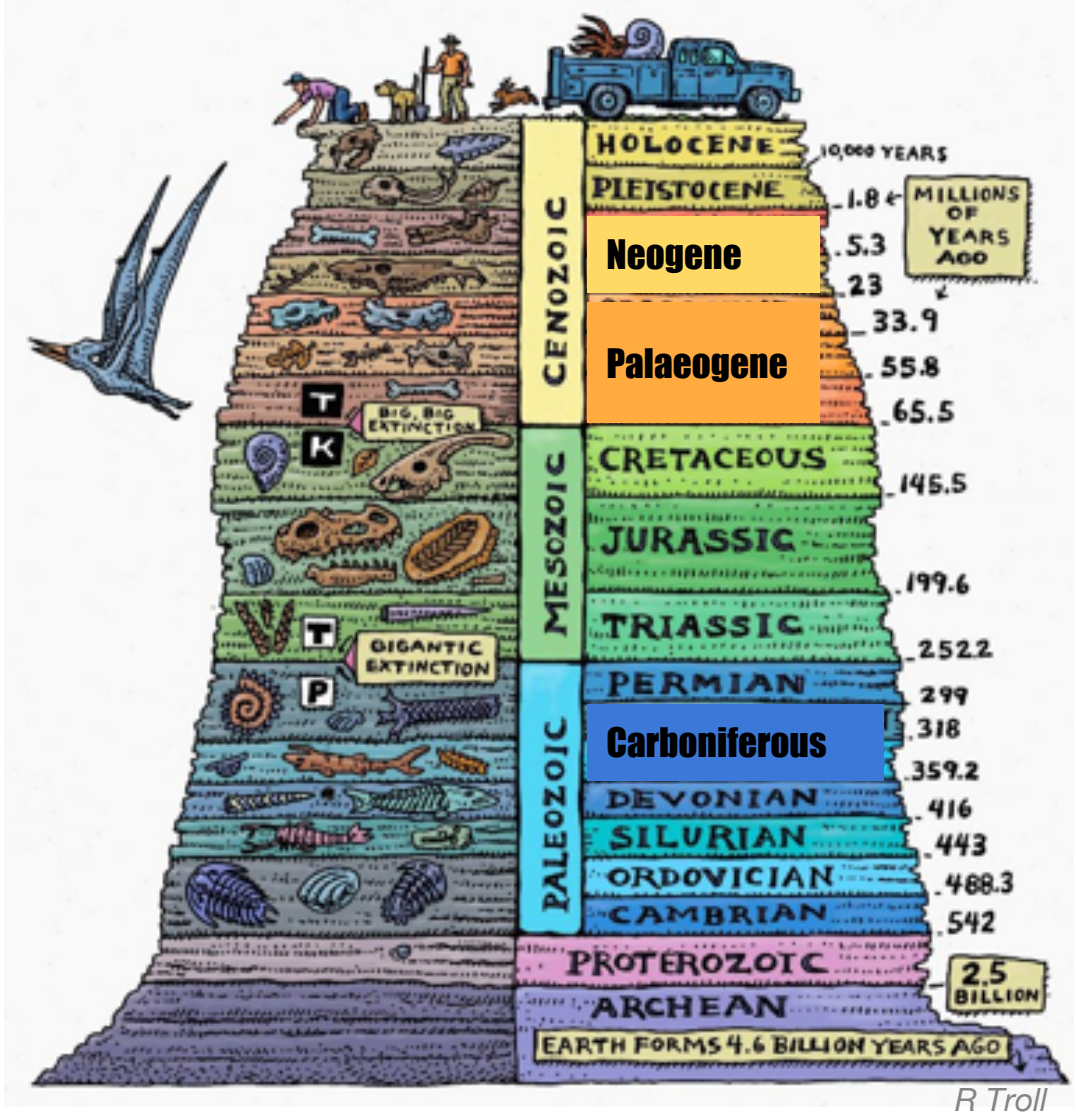
Geological Time Scale



Eons (millions of years ago):

- Hadean 4600-4000
- Archean 4000-2500
- Proterozoic 2500-540
- Phanerozoic 540-today

Geological Time Scale

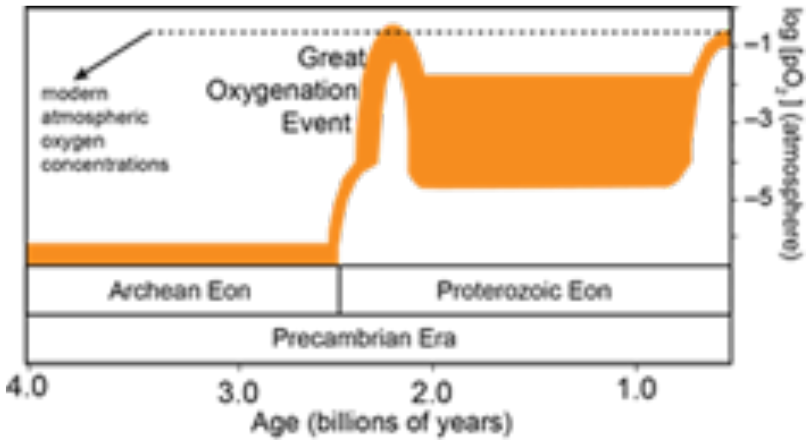


R Troll

Eons (millions of years ago):

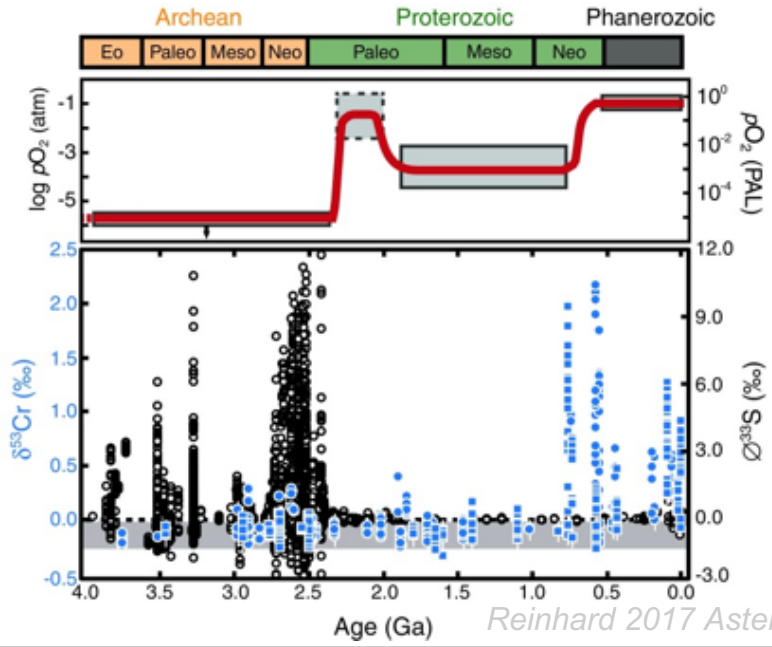
- Hadean 4600-4000
- Archean 4000-2500
- Proterozoic 2500-540
- Phanerozoic 540-today

Proterozoic Oxygen



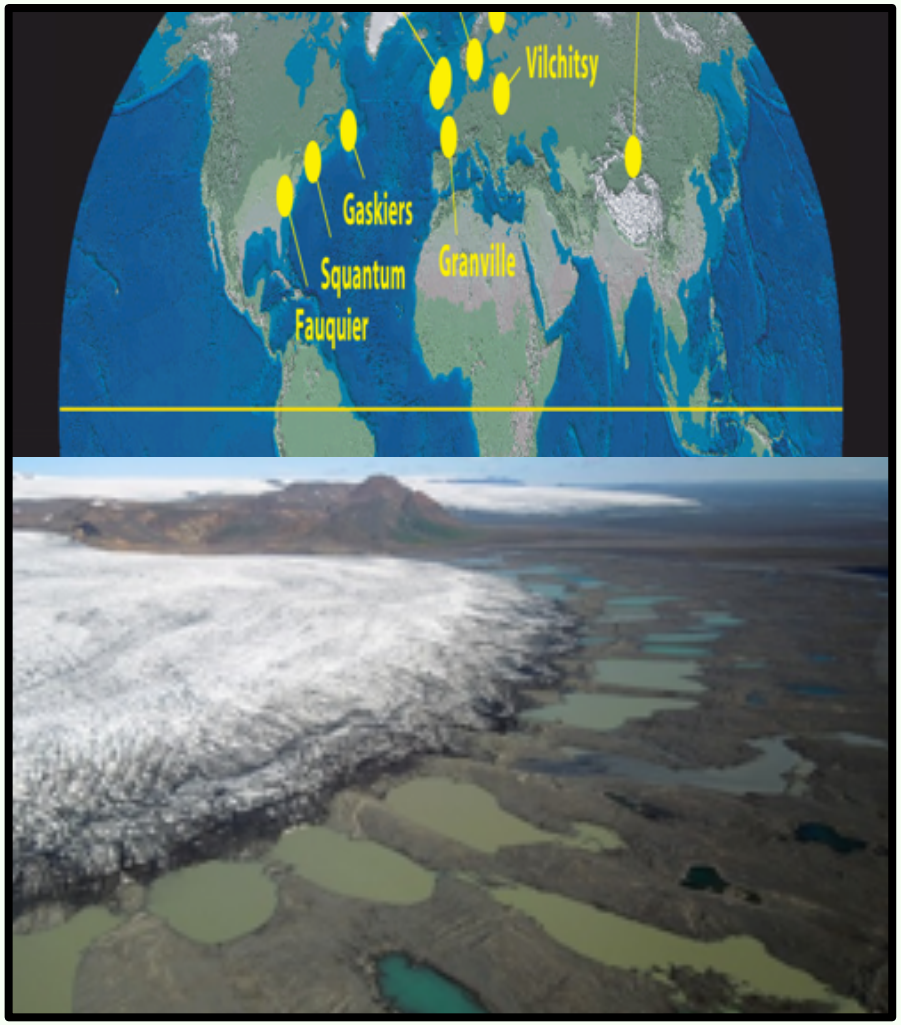
Photosynthesis lead to accumulation of atmospheric O_2 at Great Oxygenation Event

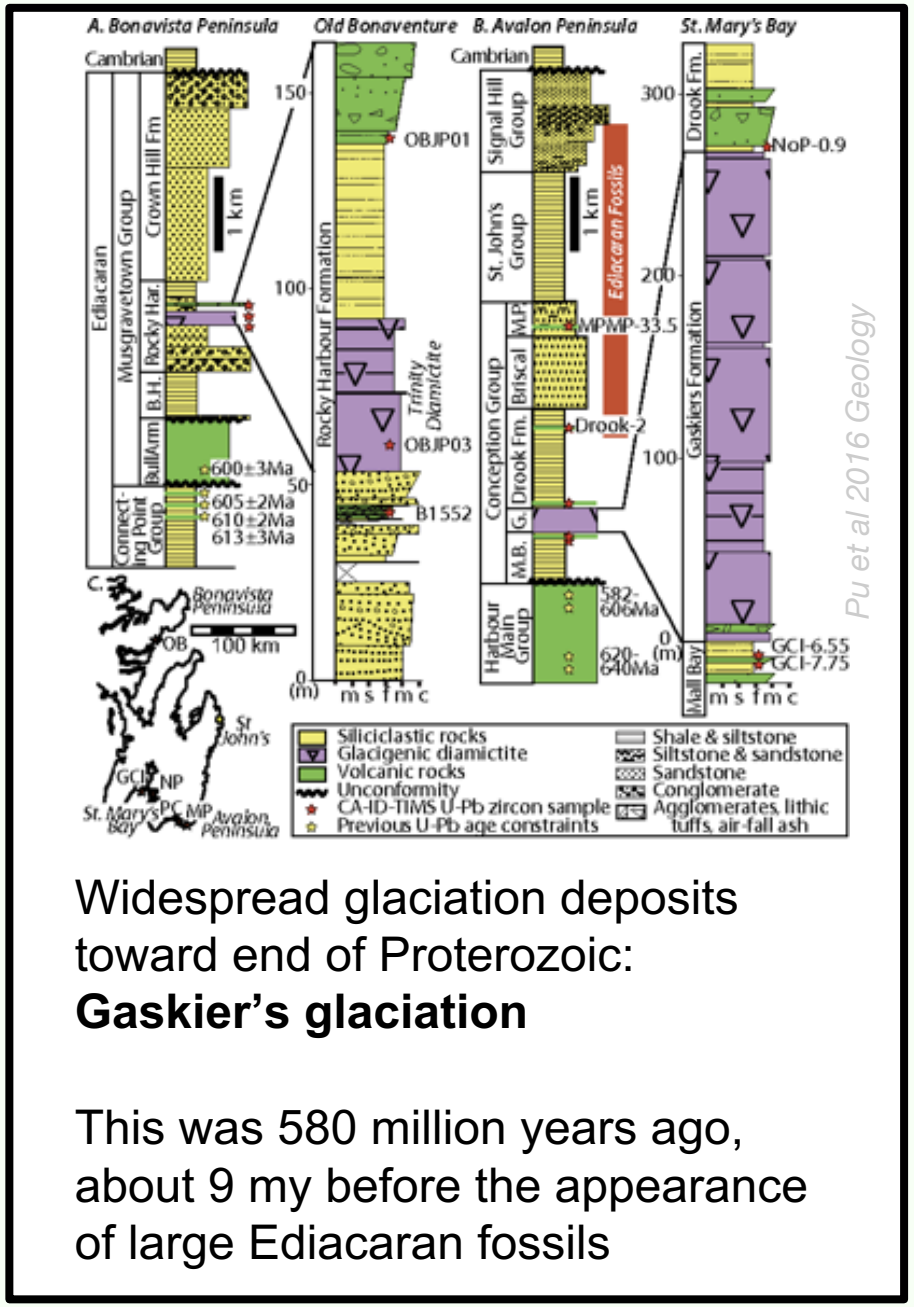
Modern levels not reached until near end of Proterozoic



Reinhard 2017 Asterobiology

Proterozoic Glaciations

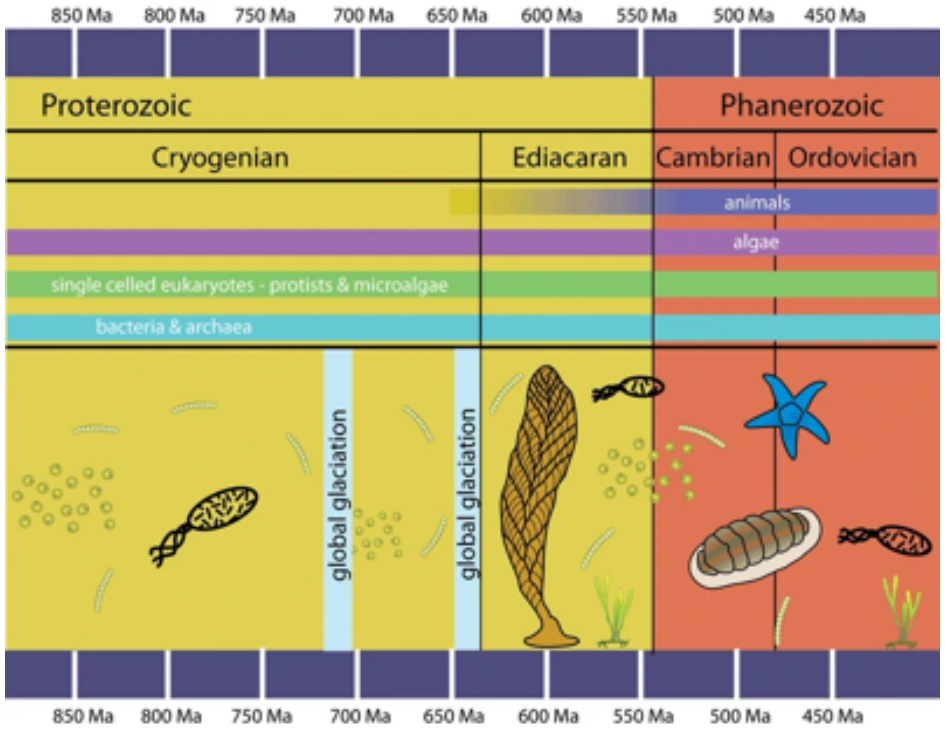




Widespread glaciation deposits toward end of Proterozoic: **Gaskier's glaciation**

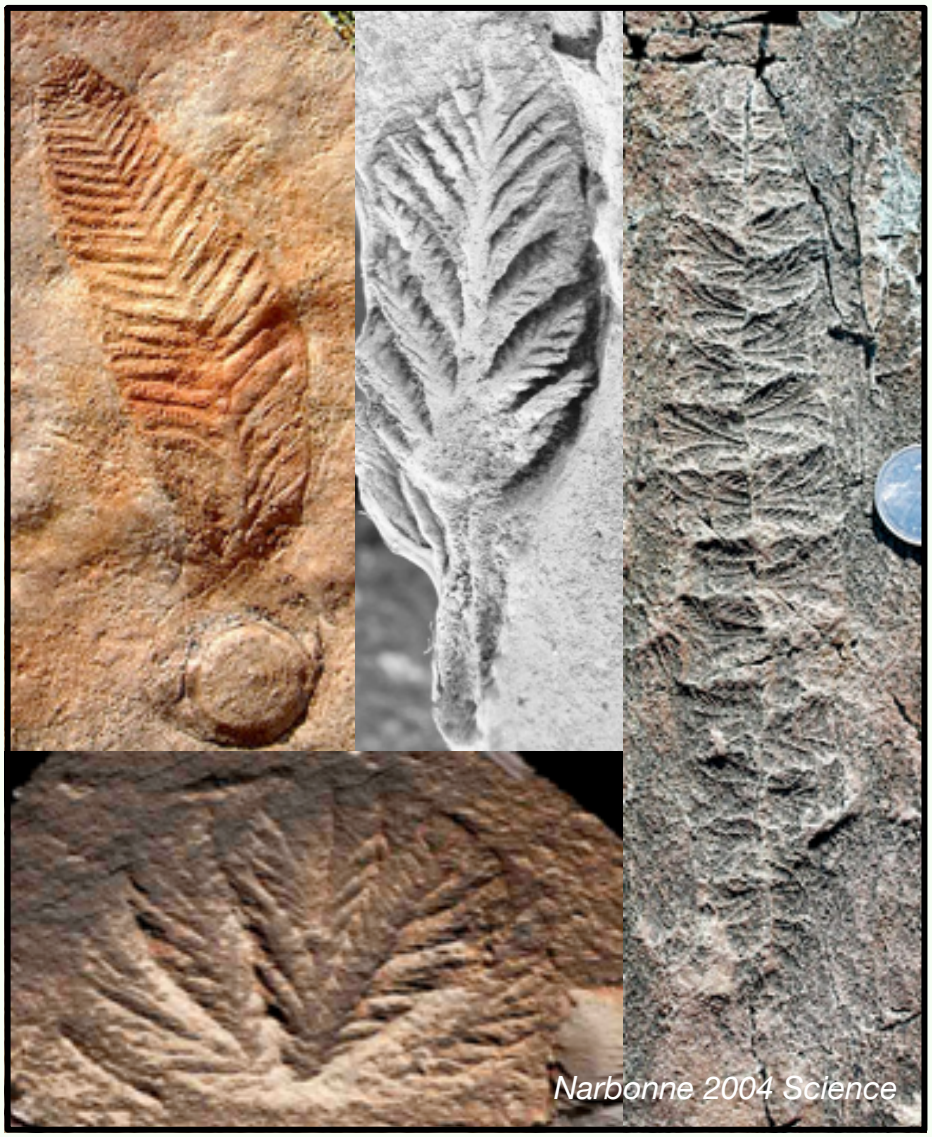
This was 580 million years ago, about 9 my before the appearance of large Ediacaran fossils

The Ediacaran



After a long period of relative stasis, large fossils turn up in the Ediacaran deposits in Newfoundland, Australia, Russia, and UK

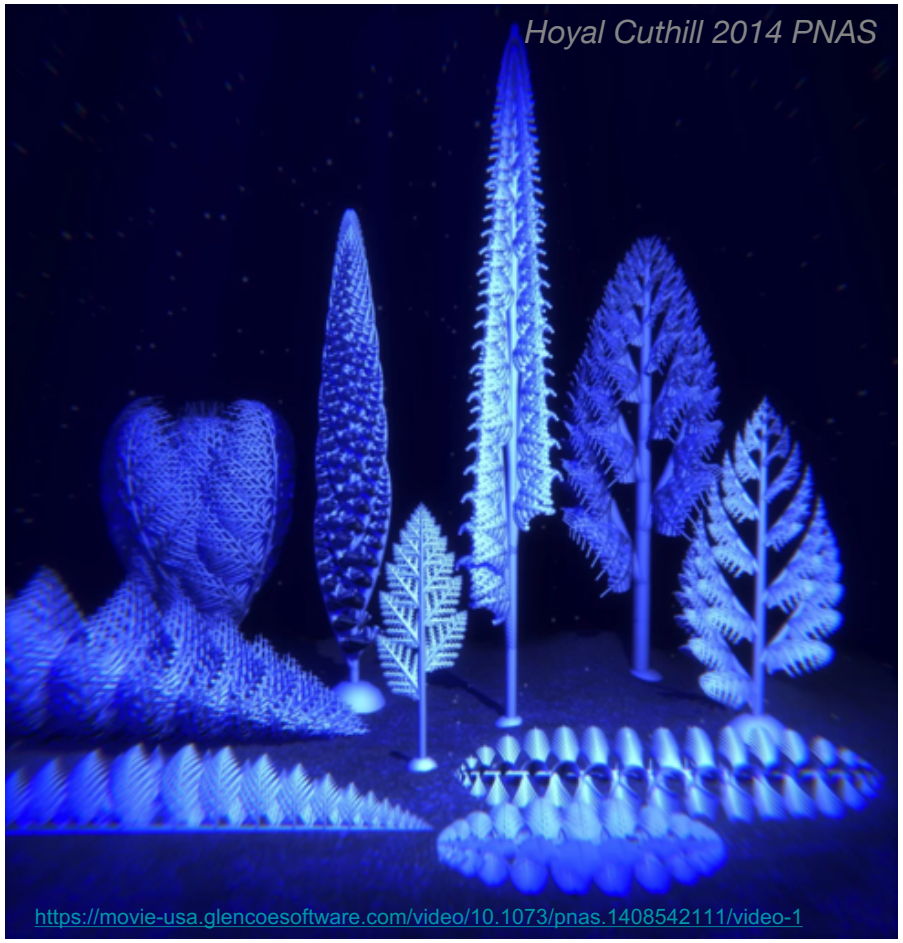
Ediacaran Fossils: “fronds”



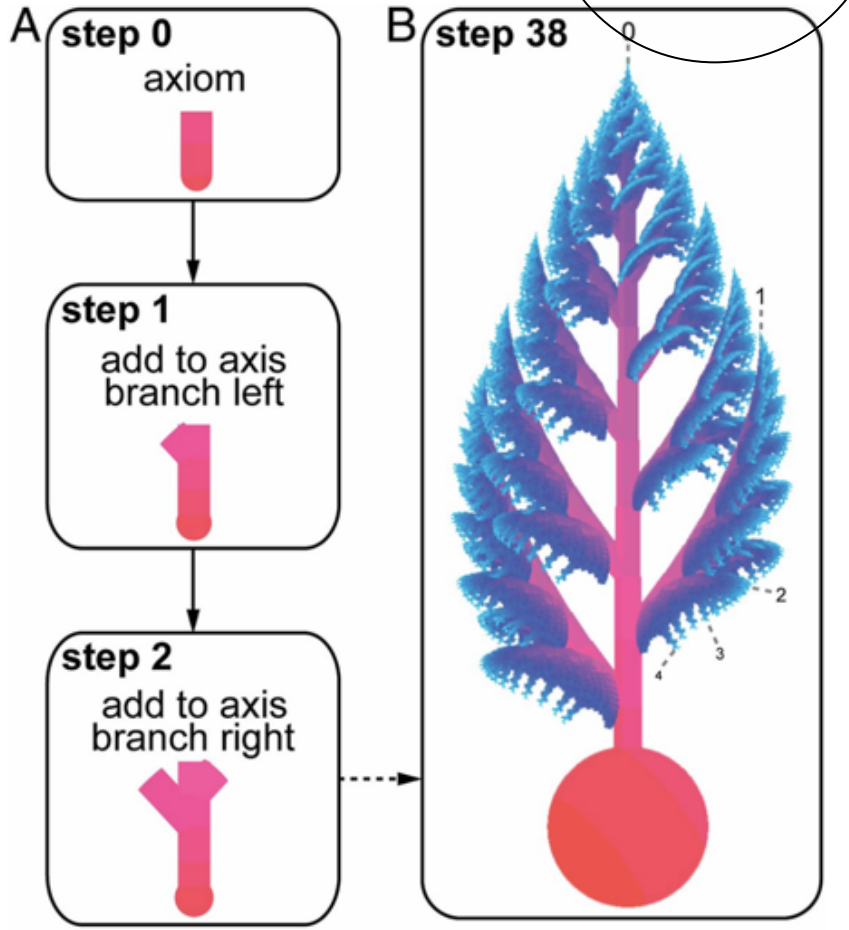
Variety of different morphologies with branching structures

Some lay flat on sea floor surface, some attach via a holdfast

Ediacaran Fossils: “fronds”



Similar fundamental morphologies, but different sizes and shapes

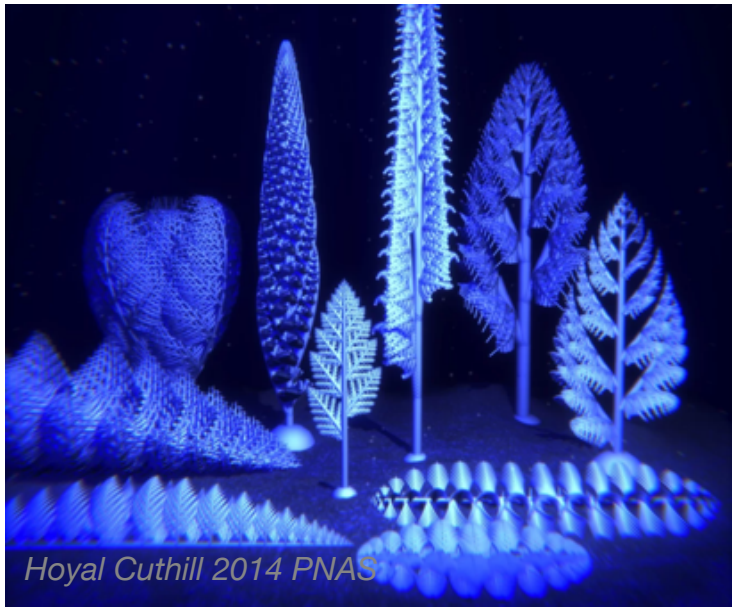


They undergo the same patterns of **fractal** growth

Ediacaran Fronds: Rangeomorphs

The frond taxa are fundamentally similar in their morphology and fractal growth, just differ slightly in their proportions and patterns

They likely form a ‘clade’:
Rangeomorphs



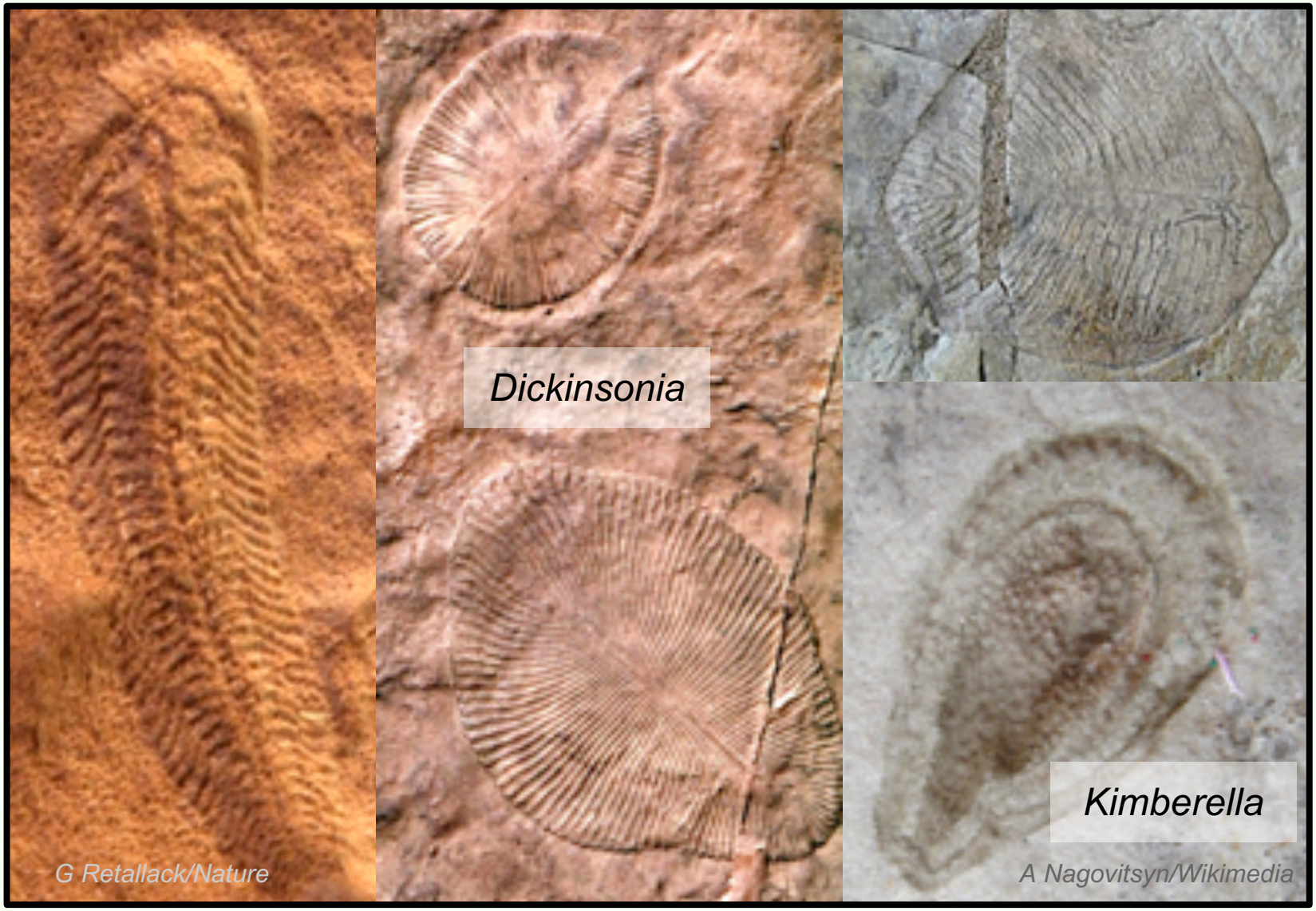
They can’t be plants
because the environment is too deep:

No light = No photosynthesis

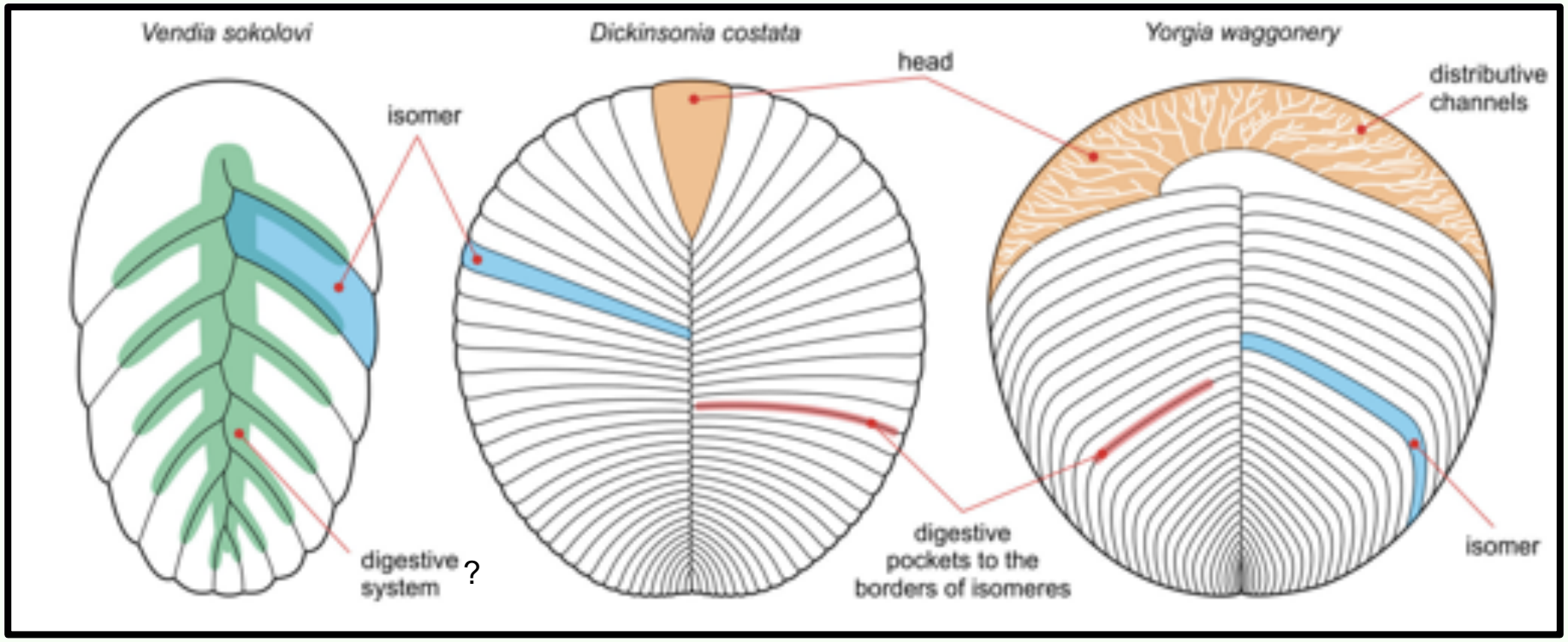
Unlikely to be proper animals given anatomy & asexual reproduction

Rangeomorphs are unique & weird!

Ediacaran Fossils: “bilaterians”



Ediacaran Fossils: “bilaterians”



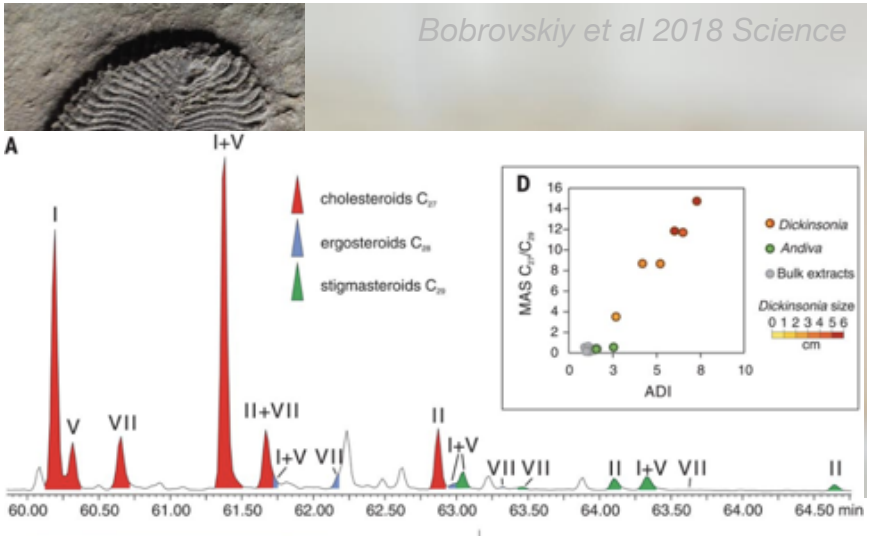
Once again, these taxa have different sizes and shapes but fundamental similarities in their constructions: a ‘head’, and a left side and a right side

But the symmetry is not classically bilaterian, it is **‘glide’ symmetry.**

Possible Ediacaran Animals?

Dickinsonia leaves traces of ‘footprints’ indicating it was digesting the underlying microbial mat

This is consistent with **osmotrophy,** as is their high surface area



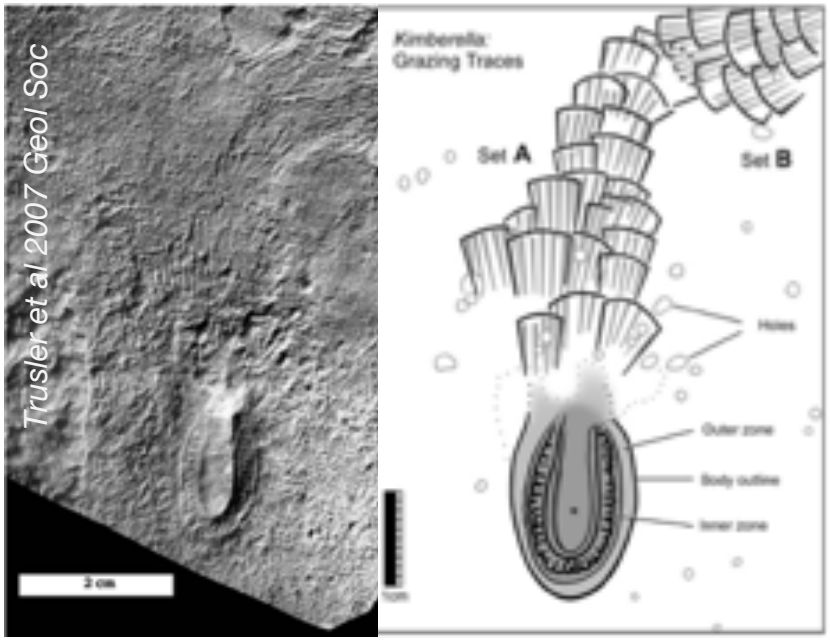
Dickinsonia has also been demonstrated to be composed of animal like molecules

Gas chromatography of organic remnants shows evidence of animal-like cholesteroids and not typically algae lipid **biomarkers**

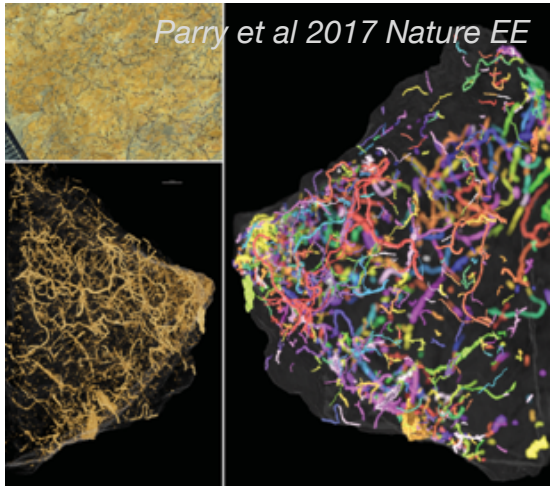
Ediacaran animal movements?

Kimberella leaves mollusc style feeding grazing traces

This makes it a convincing candidates as a proper Ediacaran animal



Some Ediacaran taxa show evidence of muscle fibres, therefore movement



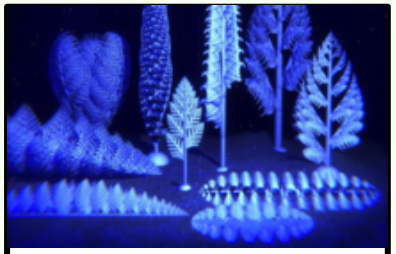
Traces of tiny burrows at the very end of the Ediacaran, possible movement

Metazoa

Fungi

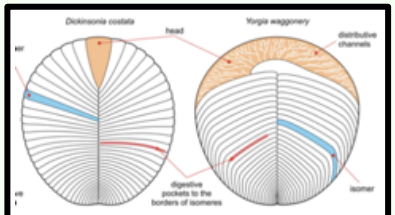
‘Simple’ Animals

Bilaterian Animals



Rangeomorphs

?



Ediacarans like *Dickinsonia*

Is *Kimberella* an advanced animal?

Origins of Animals
3.2 Ediacaran Animals?

Intended Learning Objective: Describe the Ediacaran fossil biota and appraise interpretations of their biological affinity

Question: What do you think the Ediacaran organisms were?
<10 minutes

Robert Sansom, University of Manchester

Part 3
Origin of Animals

3.3 Ediacaran Triggers

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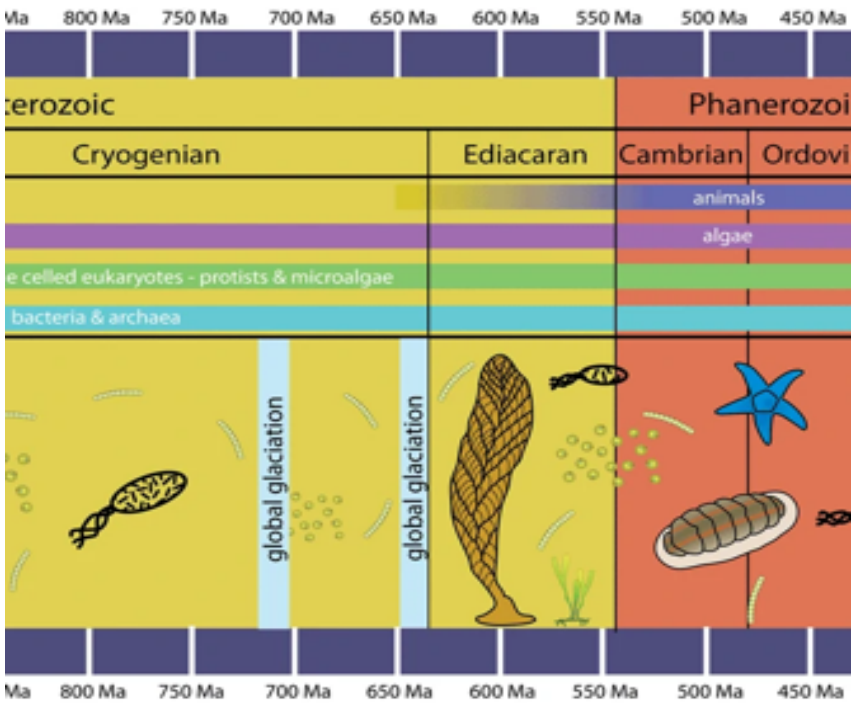
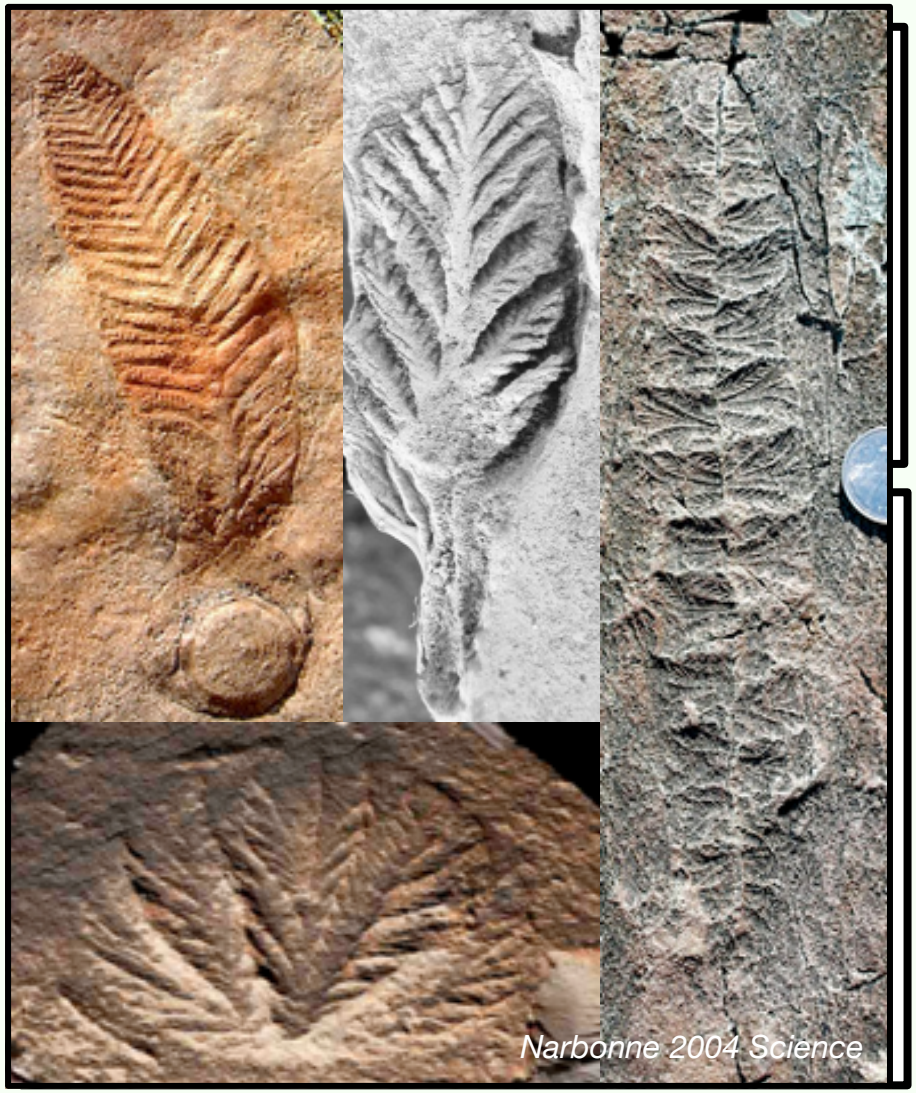
University of Manchester

Possible Causes

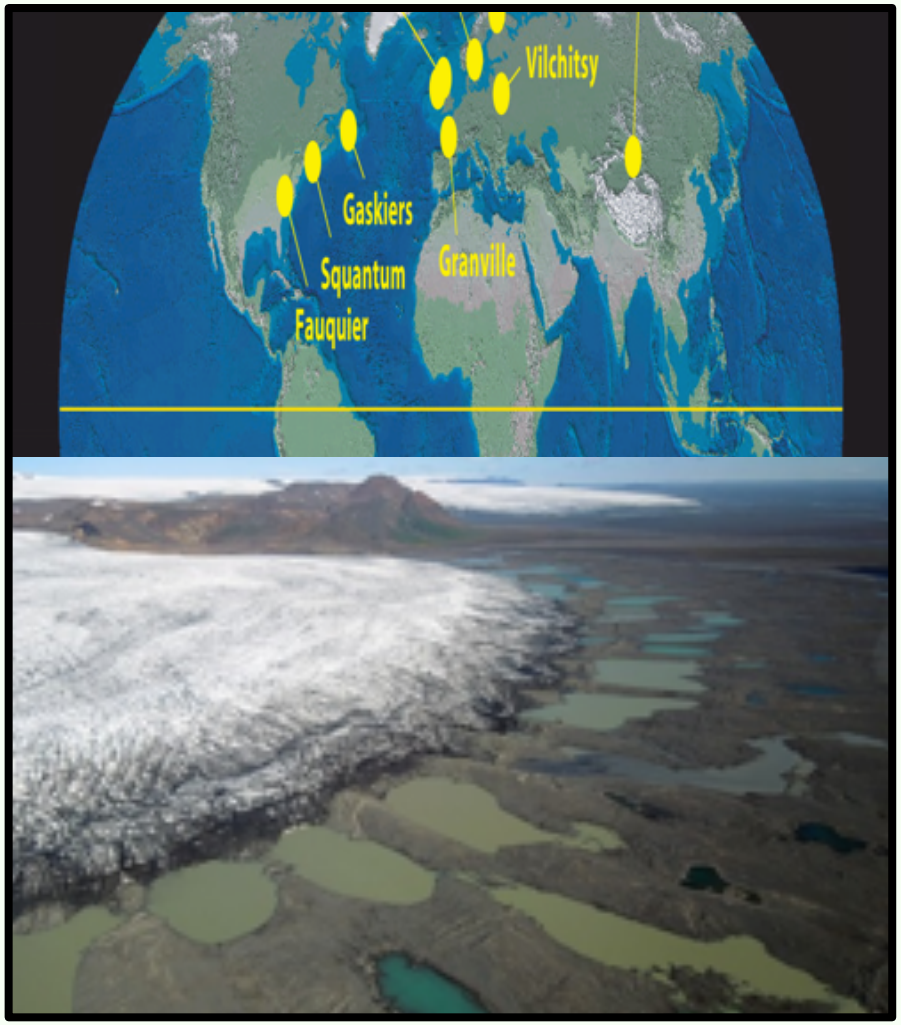


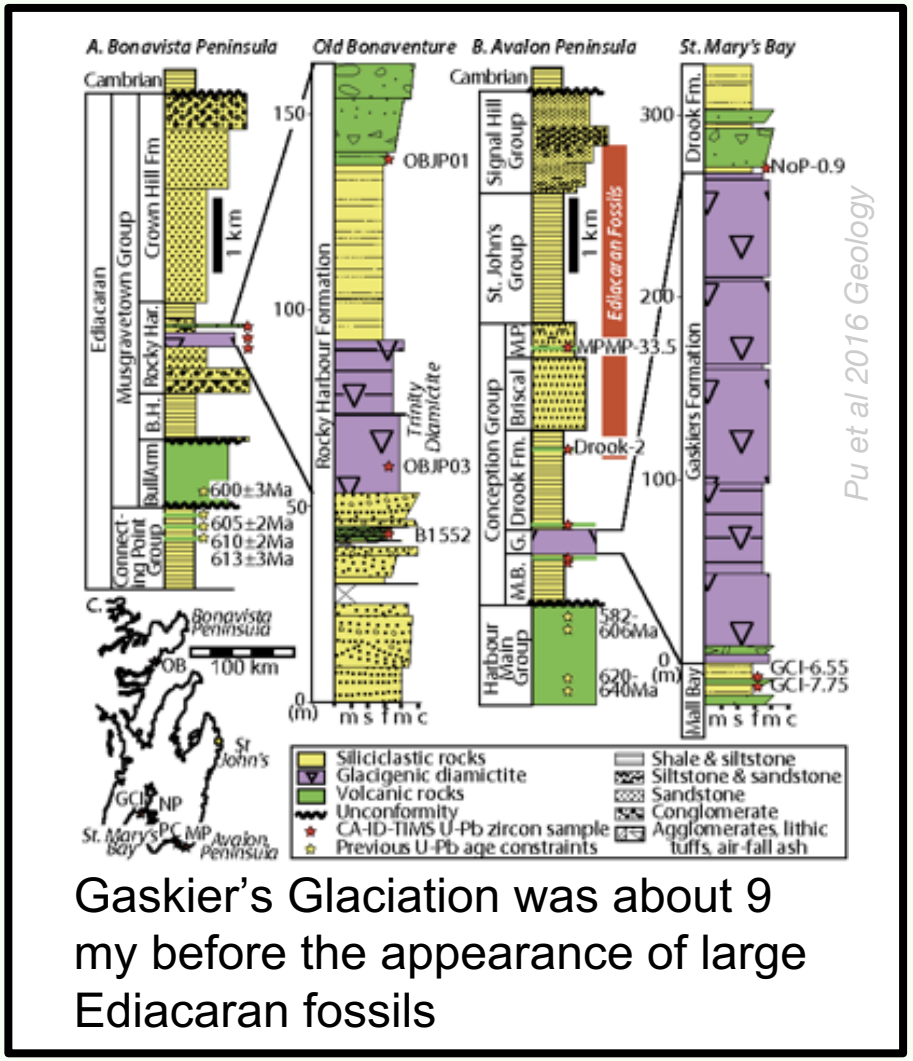
After a long period of relative stasis, large fossils turn up in the Ediacaran deposits in around the world.

Why Then?



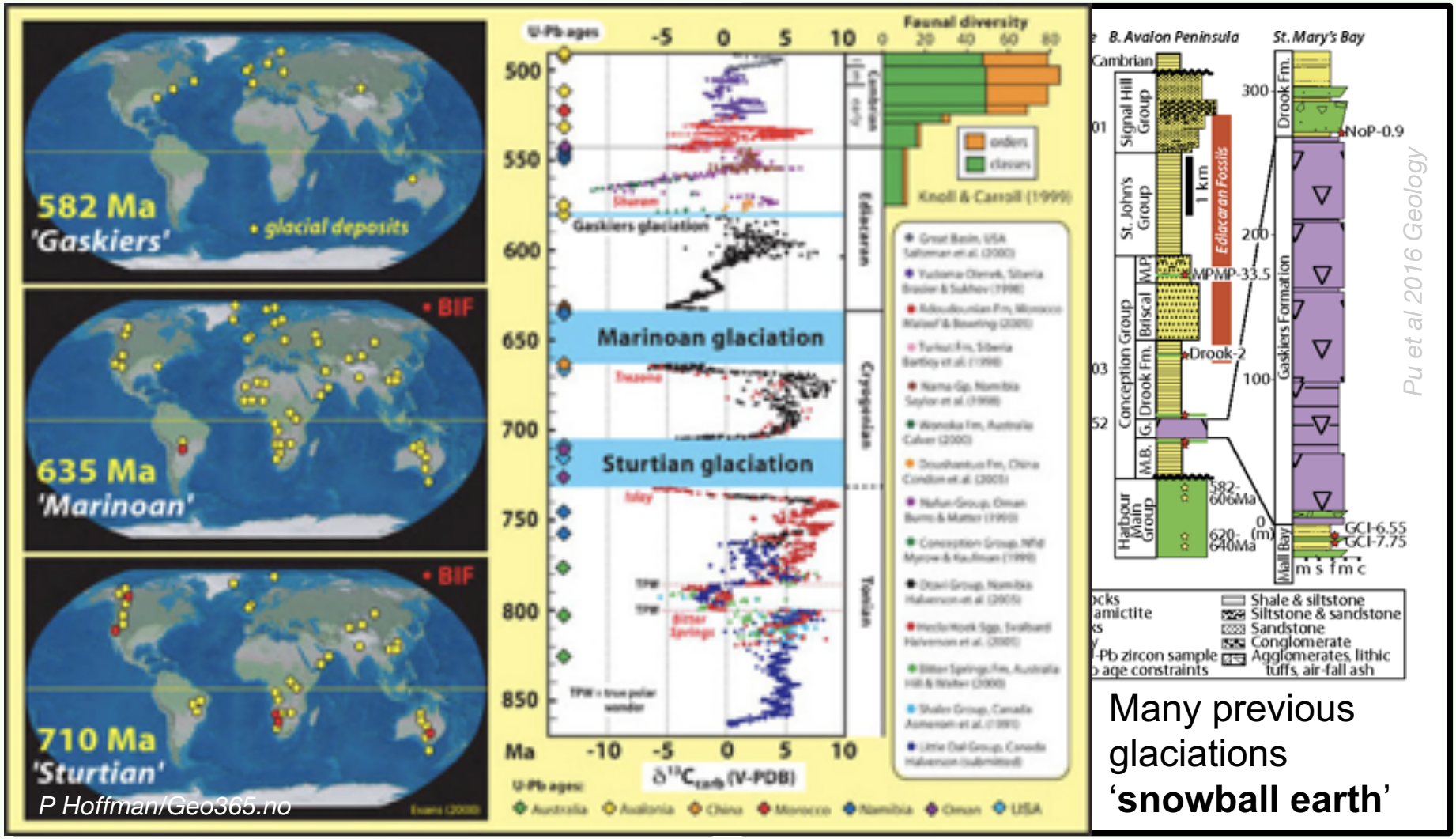
Possible Causes: Deglaciation





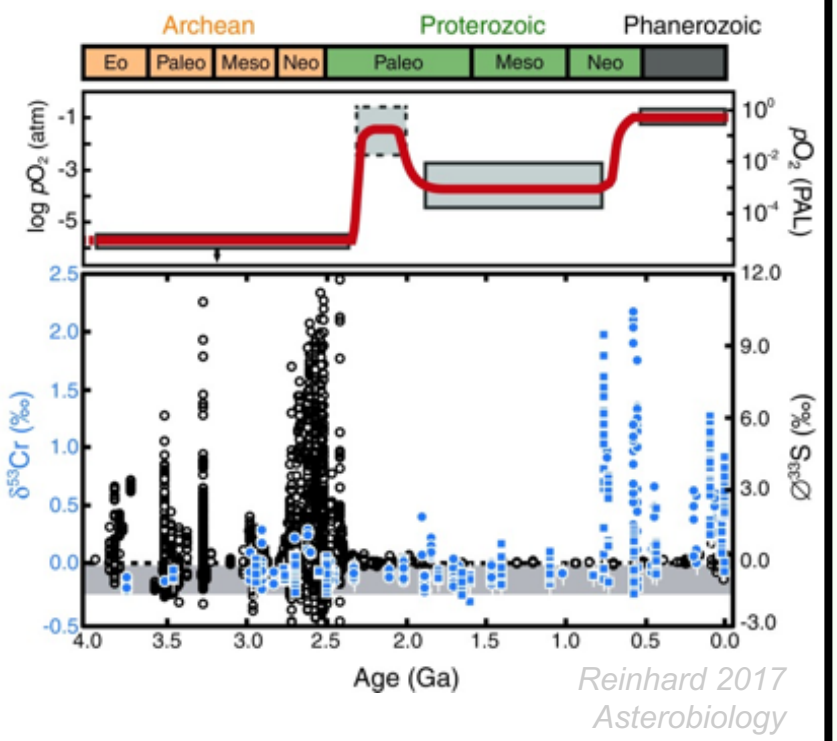
Gaskier's Glaciation was about 9 my before the appearance of large Ediacaran fossils

Possible Causes: Deglaciation



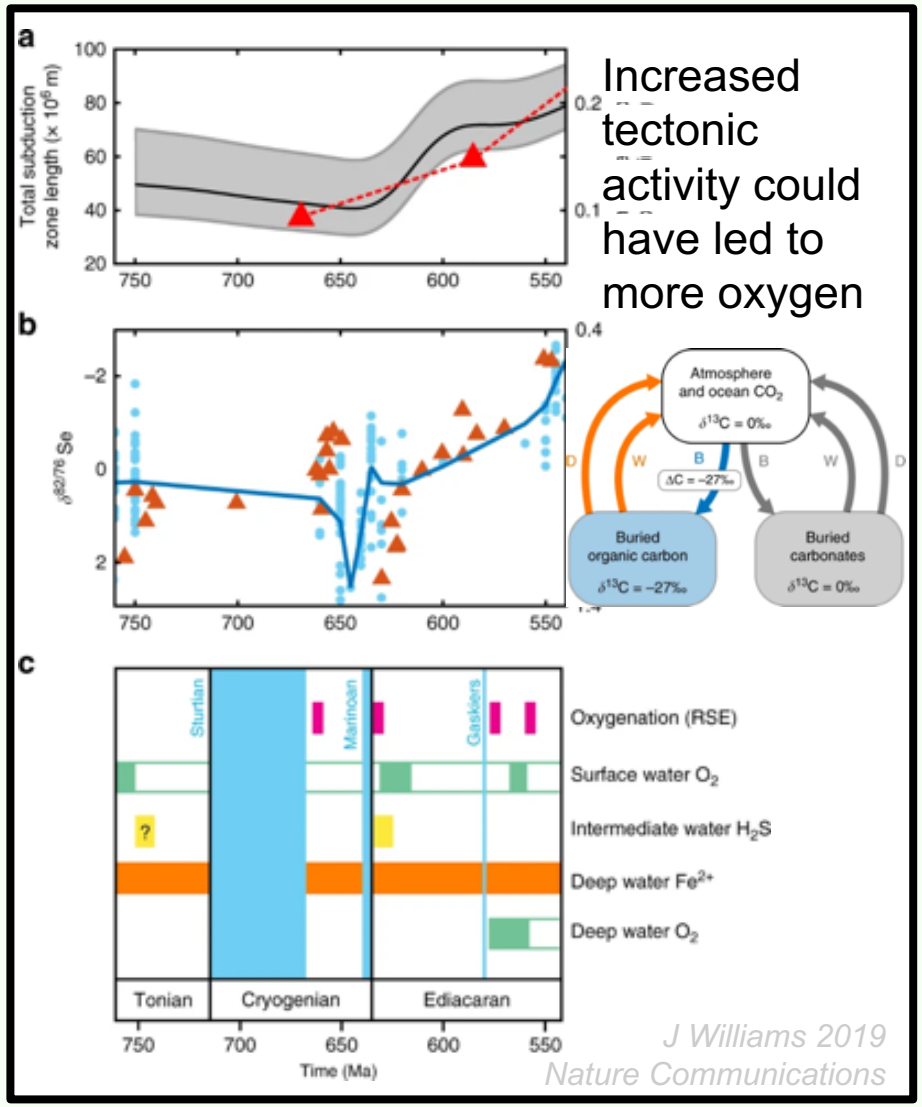
Many previous glaciations '**snowball earth**'

Possible Causes: Oxygen



Reinhard 2017
Asterobiology

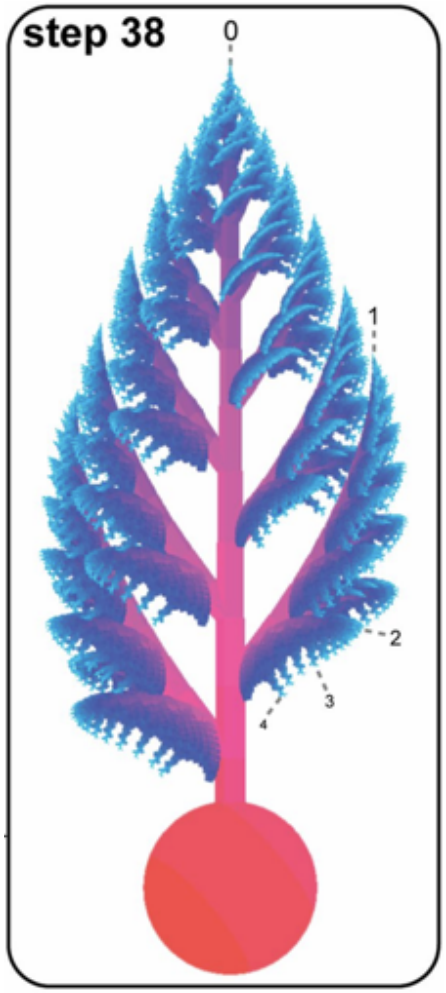
Did increasing **atmospheric** oxygen facilitate evolution of larger body sizes in the Ediacaran?



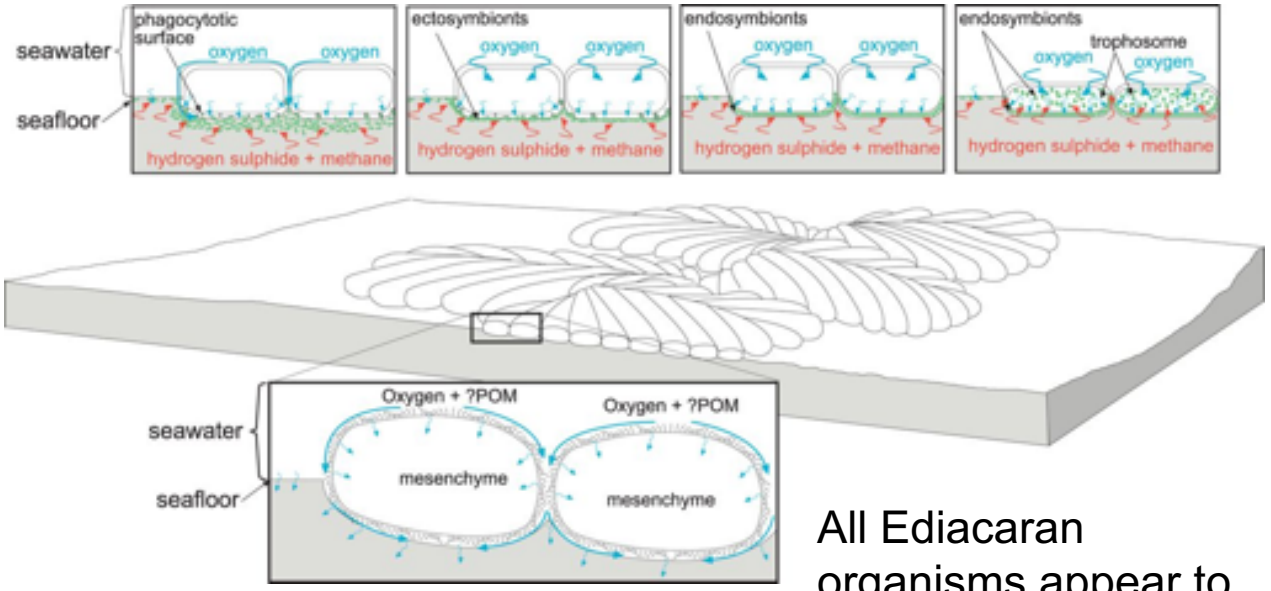
Increased tectonic activity could have led to more oxygen

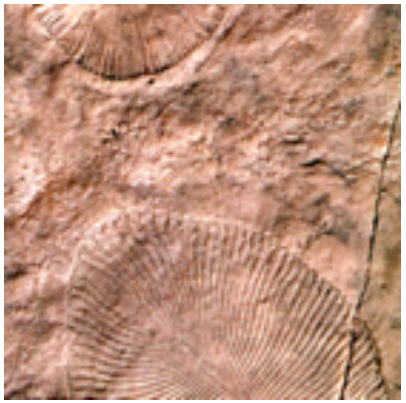
J Williams 2019
Nature Communications

Possible Causes: Oxygen



Hoyal Cuthill 2014 PNAS

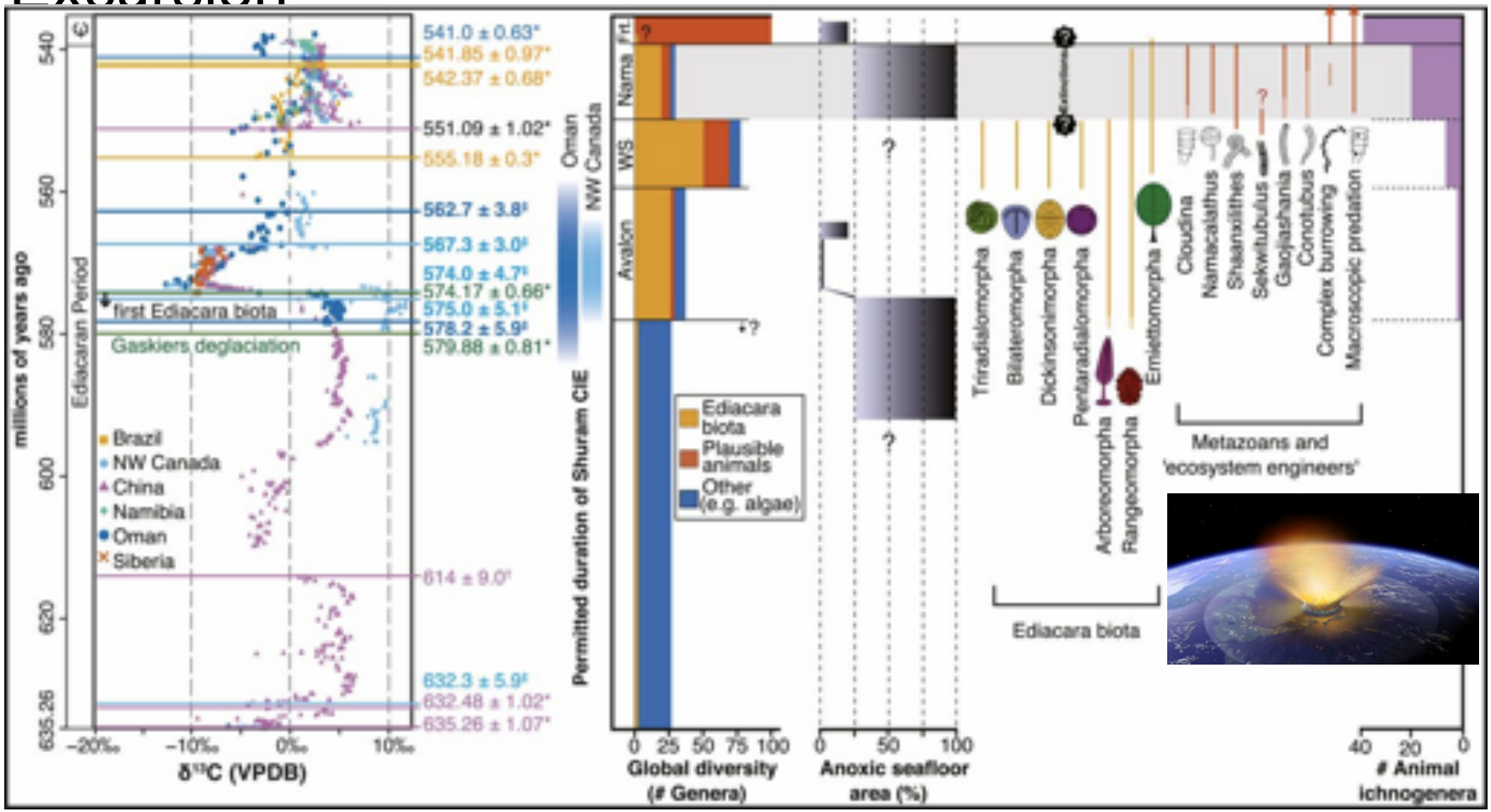




All Ediacaran organisms appear to have very high-surface area to volume ratios, important for gaseous exchange

S Dufour 2016 Geol Soc

Possible Causes: Carbon Excursion



Massive **Shuram Carbon Isotope Excursion** occurs in the Ediacaran

A Rooney 2020 PNAS

Ediacaran summary

- Life was simple for a long time after its origin
- Large organisms first appear in the Ediacaran fossil biota
- Rangeomorphs are weird fractally growing fronds and are hard to characterise
- Some Ediacaran organisms appear to be candidates for the first animals given their anatomy, composition, and behaviour

Origins of Animals
3.3 Ediacaran Evolution

Intended Learning Objective: Explain the timing and possible causes of the Ediacaran diversification in the context of earth history

Question: Why did Ediacaran biota first evolve then?
Shower thought

Robert Sansom, University of Manchester

Part 3
Origins of Animals

3.4 Evolutionary Trees

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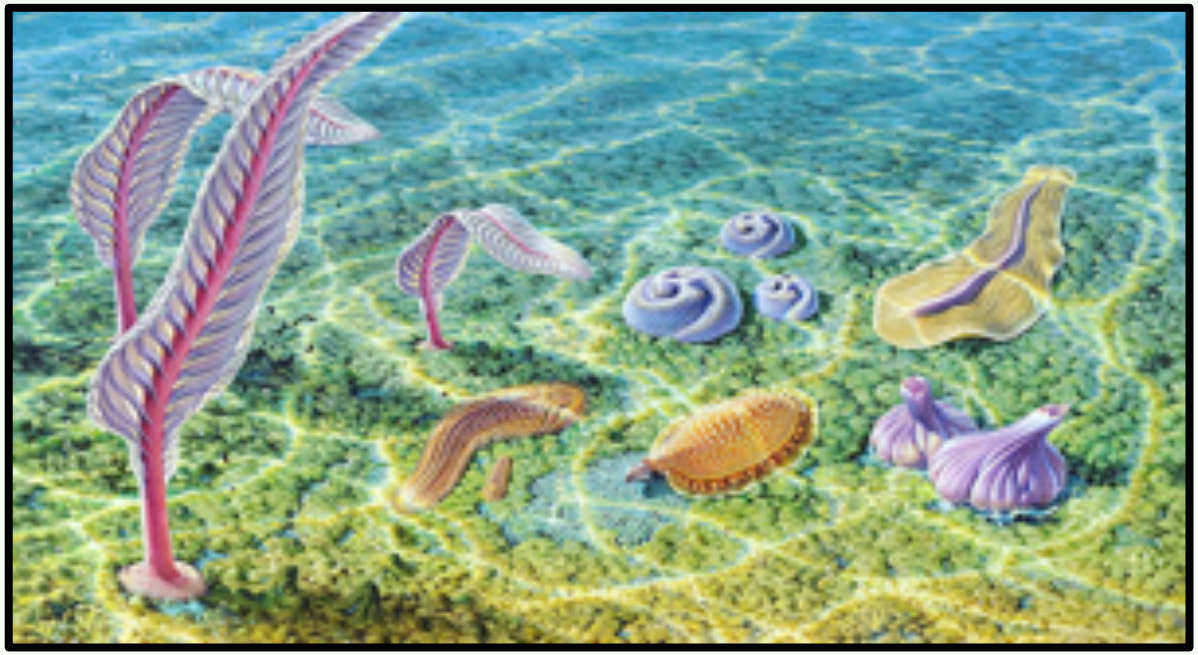
Robert Sansom
University of Manchester



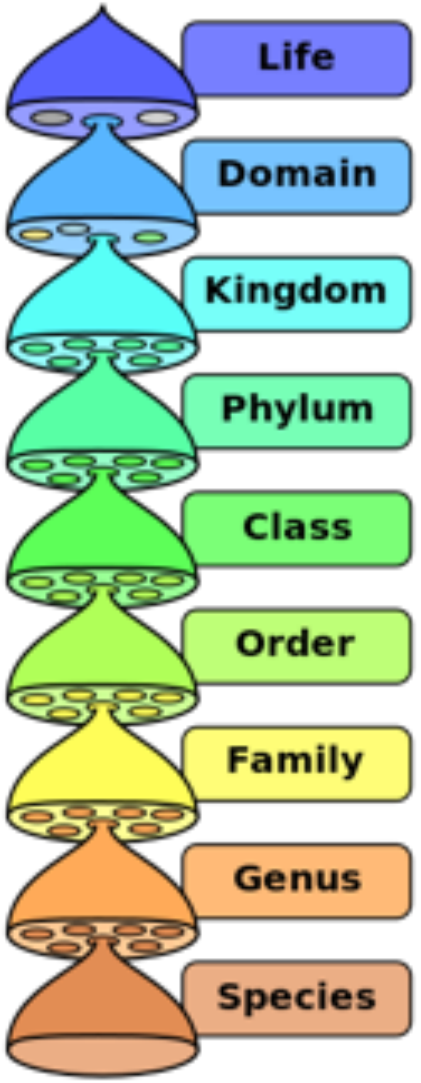
Ediacaran ‘proto’ animals

Complex modern life

Classification of Animals

Bilaterians: they have a left/right sides, head and tail, top and bottom, and gut from mouth to anus. There are many phyla of bilateria

Taxonomy: *Hierarchical Classification*



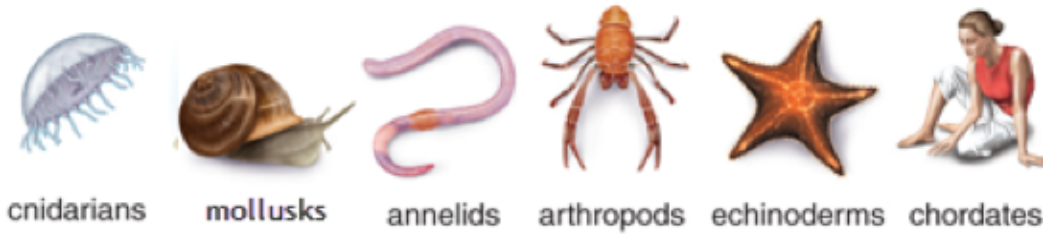
Animal Phyla

A **phylum** is a unique body plan

There are around 30 modern animal phyla

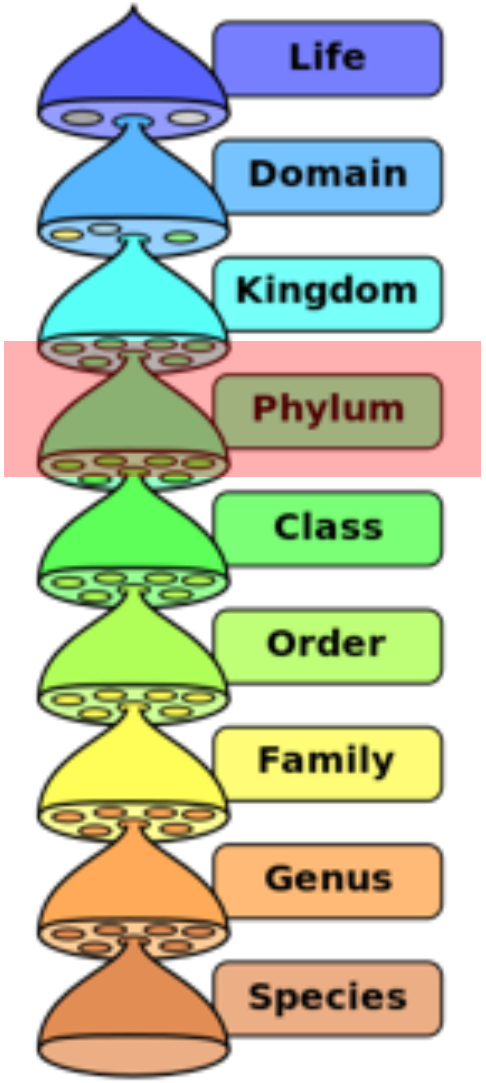
Most are bilaterians

Some major phyla



Some minor phyla

Taxonomy: *Hierarchical Classification*



Animal Phyla

A **phylum** is a unique body plan e.g.

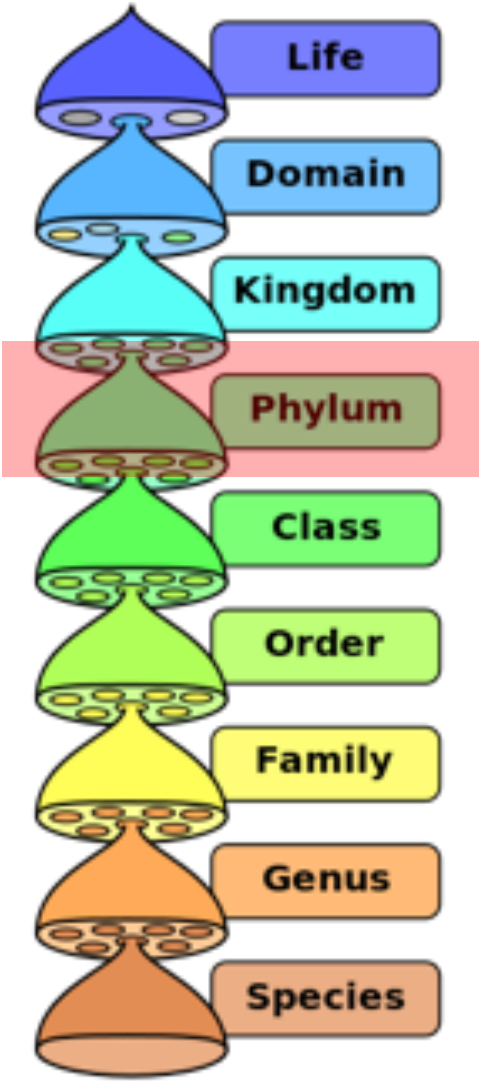
Arthropoda:
Bilateria with jointed appendages

Insects, crustaceans, spiders, scorpions, millipedes

Chordata:
Bilateria with notochord and dorsal nerve cord

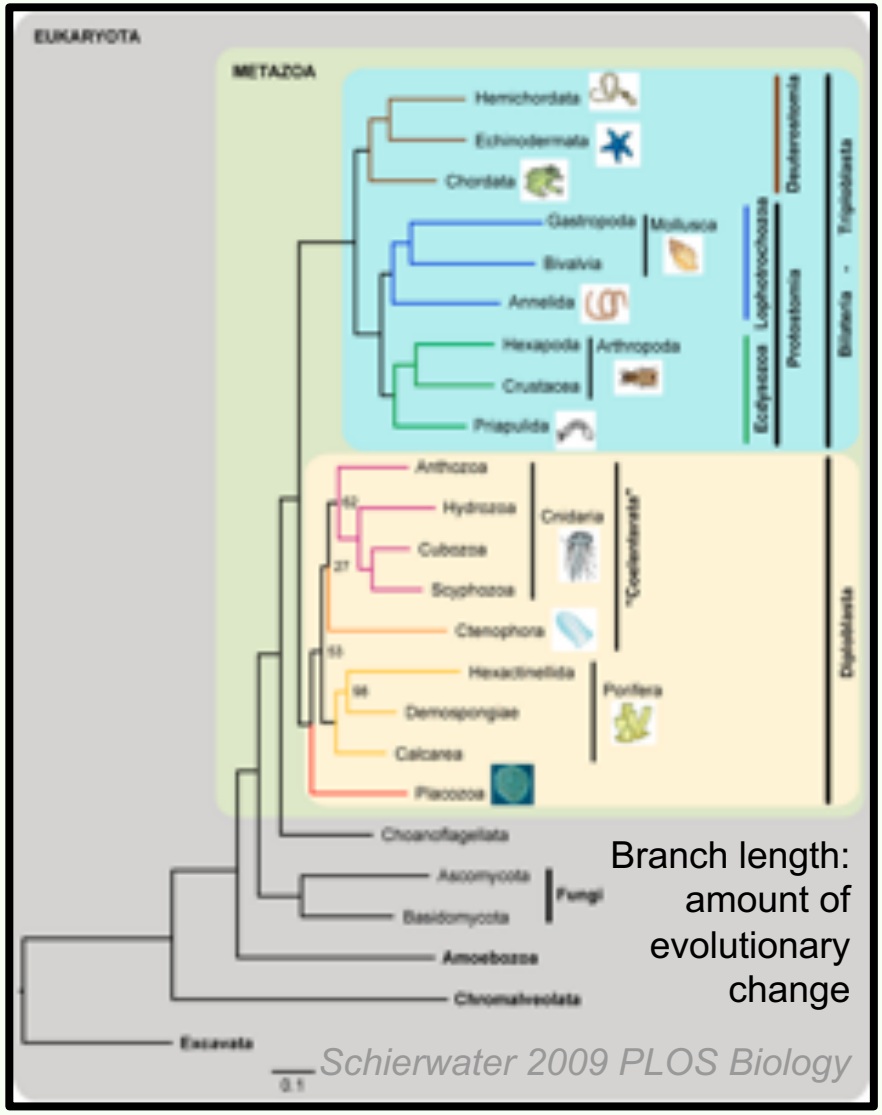
Tunicates, lancelets and all vertebrates

Taxonomy: *Hierarchical Classification*



Metazoan Phylogeny

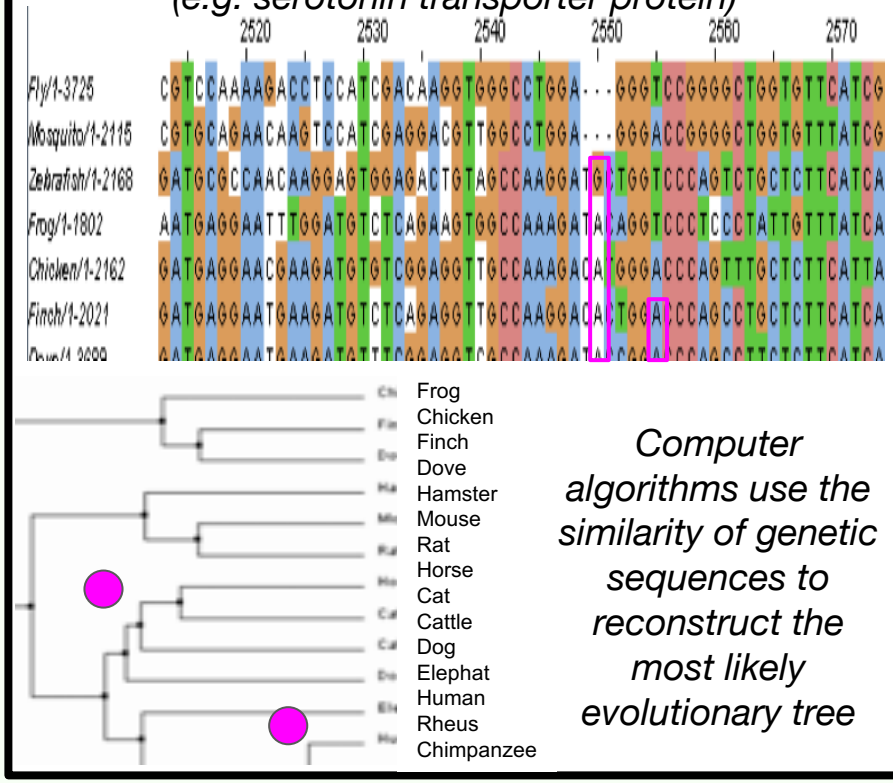
A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



Branch length: amount of evolutionary change

This is a **molecular phylogeny** - it is based on the similarities and differences in the genetic sequences

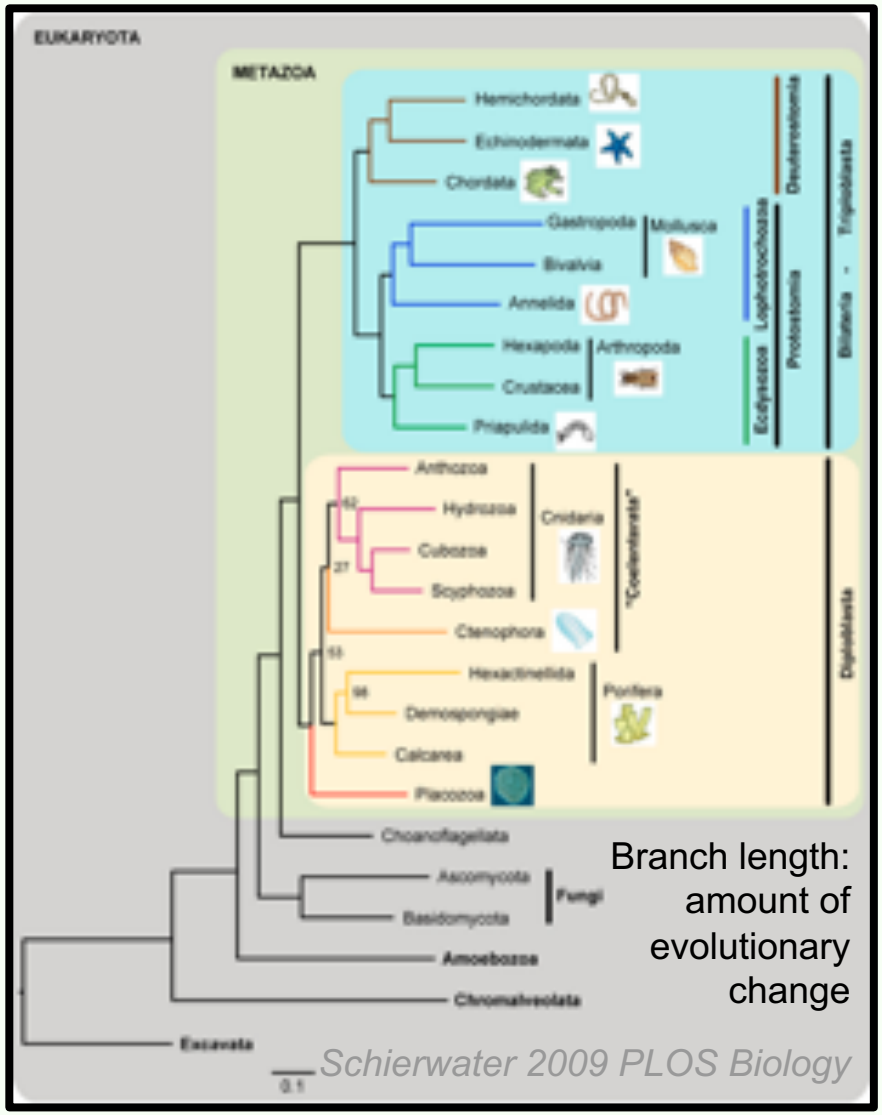
Genetic sequences of genes or whole genome (e.g. serotonin transporter protein)



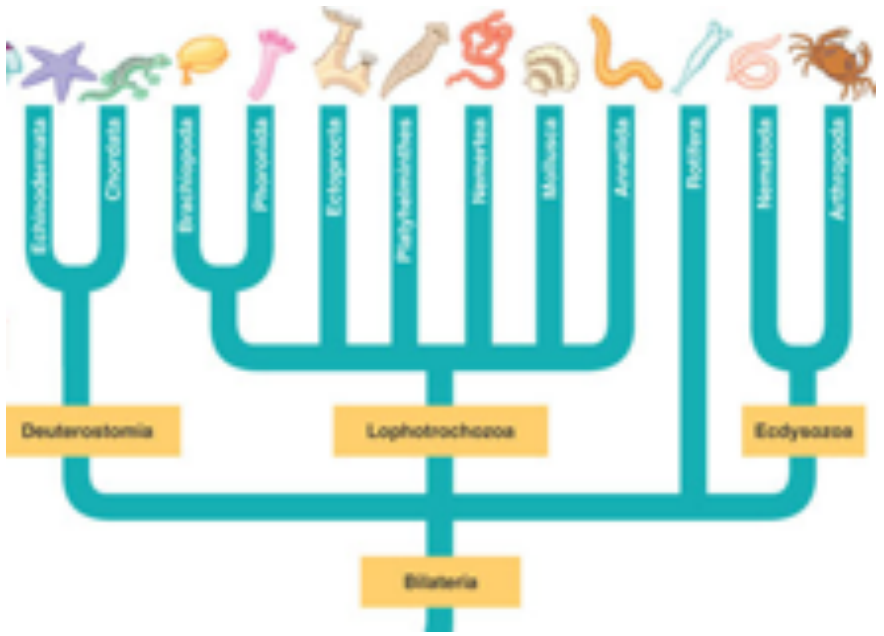
Computer algorithms use the similarity of genetic sequences to reconstruct the most likely evolutionary tree

Metazoan Phylogeny

A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



Branch length: amount of evolutionary change

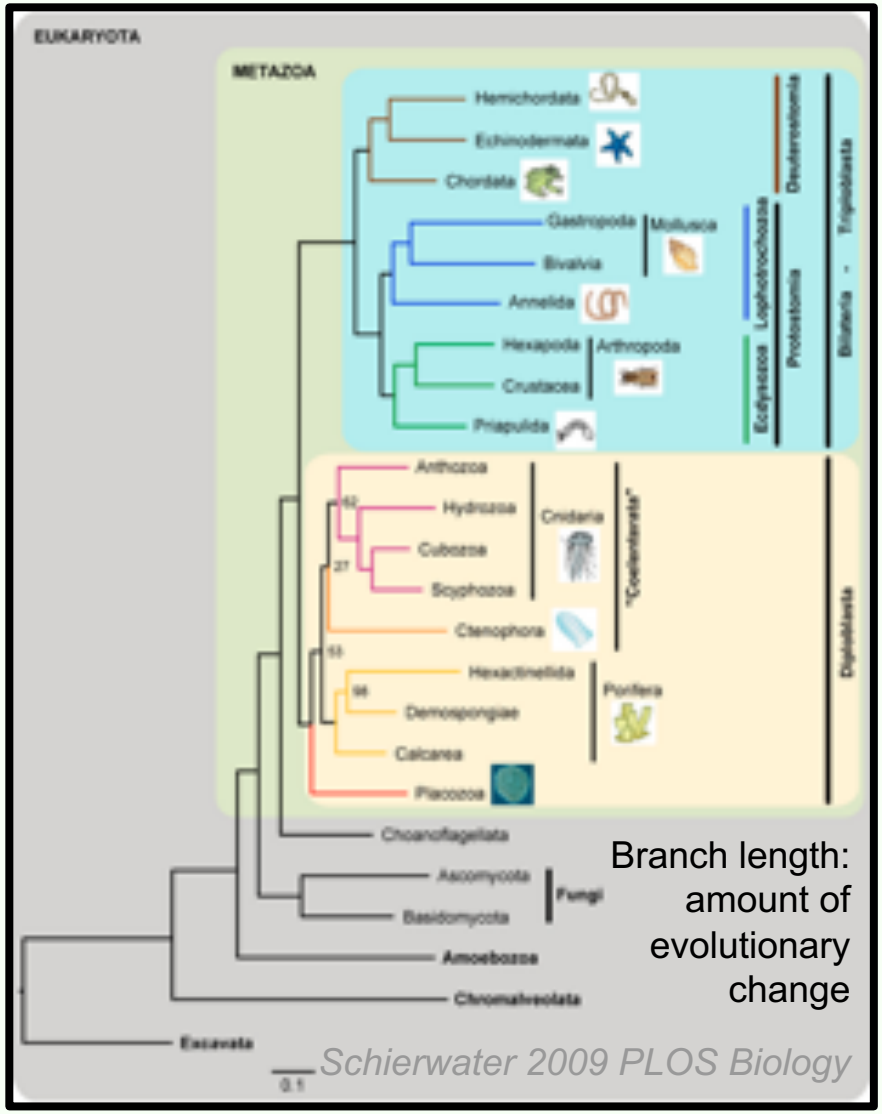


There are three **superphyla** of bilaterian animals (groupings of phyla):

- **Lophotrochozoa** (molluscs, annelids and lots of other phyla)
- **Ecdysozoa** (arthropods & allied phyla)
- **Deuterostomes** (chordates inc. vertebrates, and echinoderms)

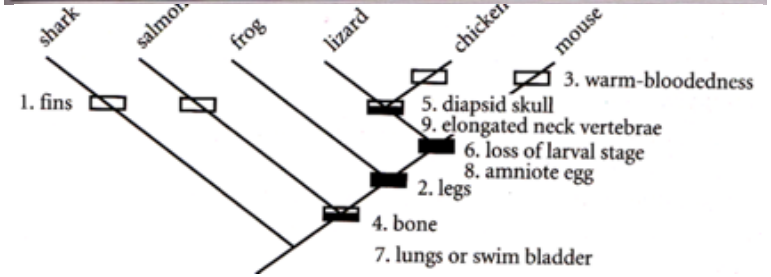
Metazoan Phylogeny

A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



Ancient fossils don't have DNA, so we need to use **morphological** data

Character	Shark	Salmon	Frog	Lizard	Chicken	Mouse
1 Fins	1	1	0	0	0	0
2 Legs	0	0	1	1	1	1
3 Warm-bloodedness	0	0	0	0	1	1
4 Bone	0	1	1	1	1	1
5 Diapsid skull	0	0	0	1	1	0
6 Loss of larval stage	0	0	0	1	1	1
7 Lung or swim bladder	0	1	1	1	1	1
8 Amniote egg	0	0	0	1	1	1
9 Elongated neck vertebrae	0	0	0	1	1	0
10 Marginal teeth	0	1	1	1	1	1



Anatomy broken down into binary characters (e.g. presence/absence), used to build relationships

Origins of Animals
3.4 Phylogeny

Intended Learning Objective: Explain how to reconstruct evolutionary relationships from molecular and morphological data using phylogenetics

Question: Can you find a published phylogeny of any group, living or dead (and tell us how it was built)?
<15 minutes

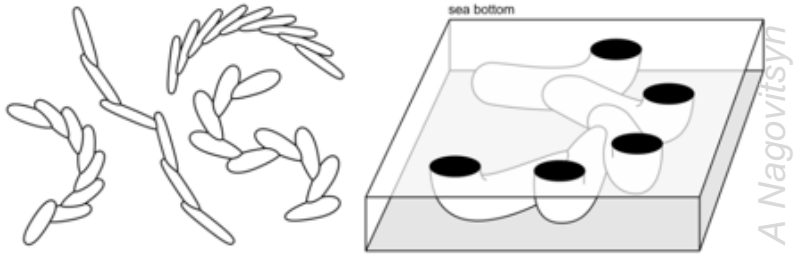
Robert Sansom, University of Manchester

Part 3
The Origins of Animals

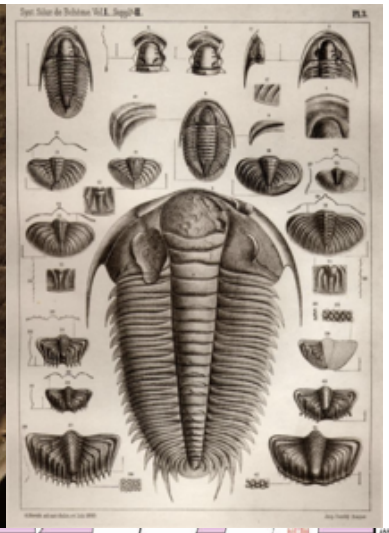
3.5 The Burgess Shale

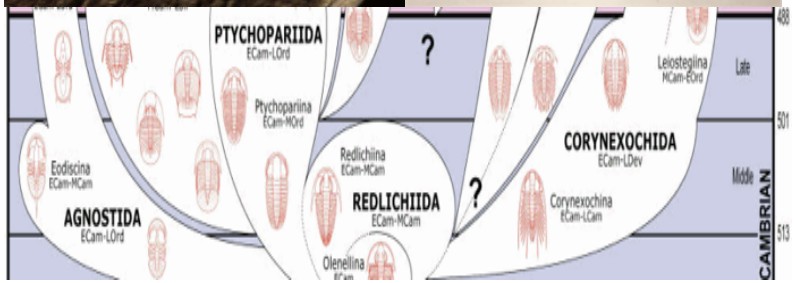
EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Beginning of the Cambrian



The trace fossil *Treptichnus* (a worm burrow) defines the beginning of the Cambrian

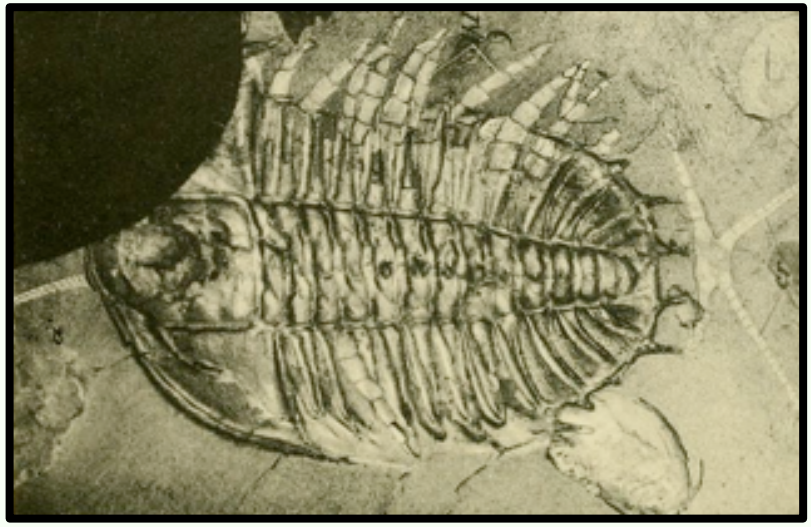




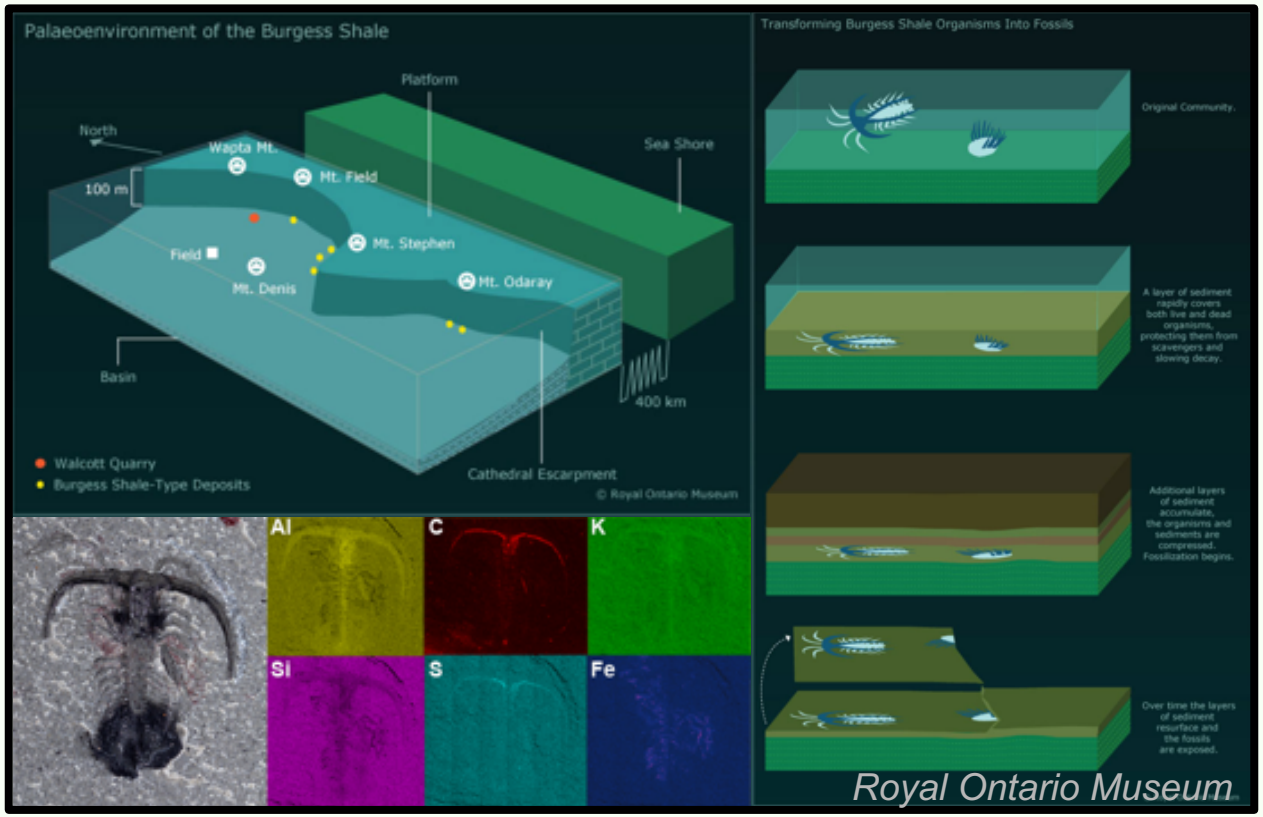
Trilobites, fossil arthropods, are suddenly abundant and diverse in the Cambrian

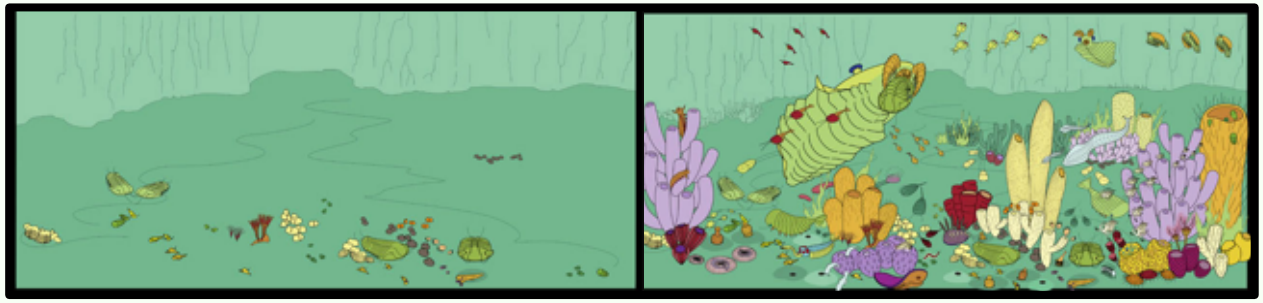
The Burgess Shale

A middle Cambrian fossil deposit in Canada with exceptionally soft tissue preservation



The Burgess Shale: Preservation





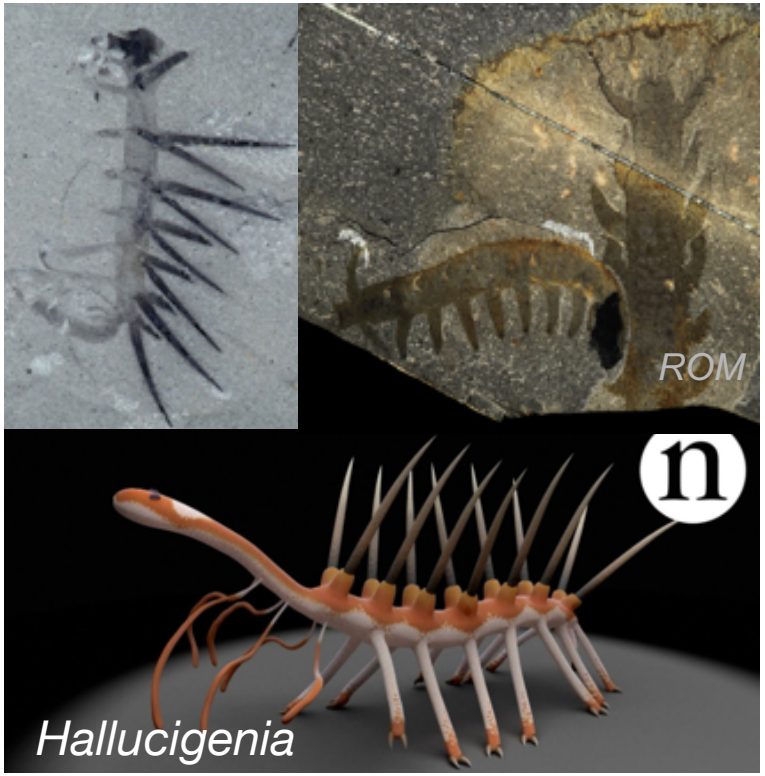
The marine community was buried rapidly following collapse of a sea cliff

Sediment accumulated and underwent **diagenesis** preserving soft tissues as clays

Whole community preserved, not just shells and hard bits

The Burgess Shale: Fossil taxa

Large compound eyes, raptorial appendages and swimming flaps



Worms with stubby walking legs and very unusual anatomies

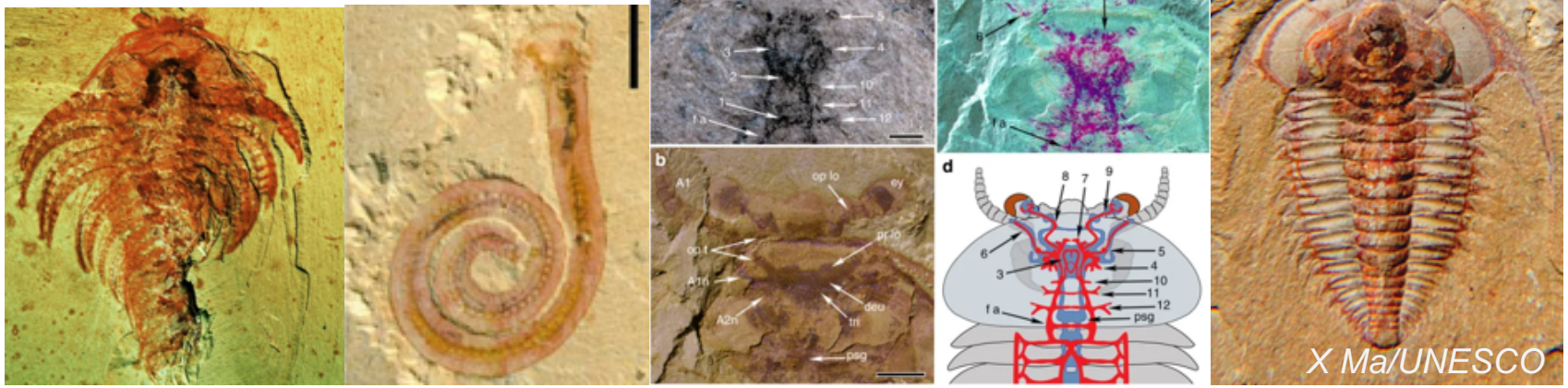
The Burgess Shale: Fossil taxa

Sessile, filter feeding organisms with front and back ‘shells’

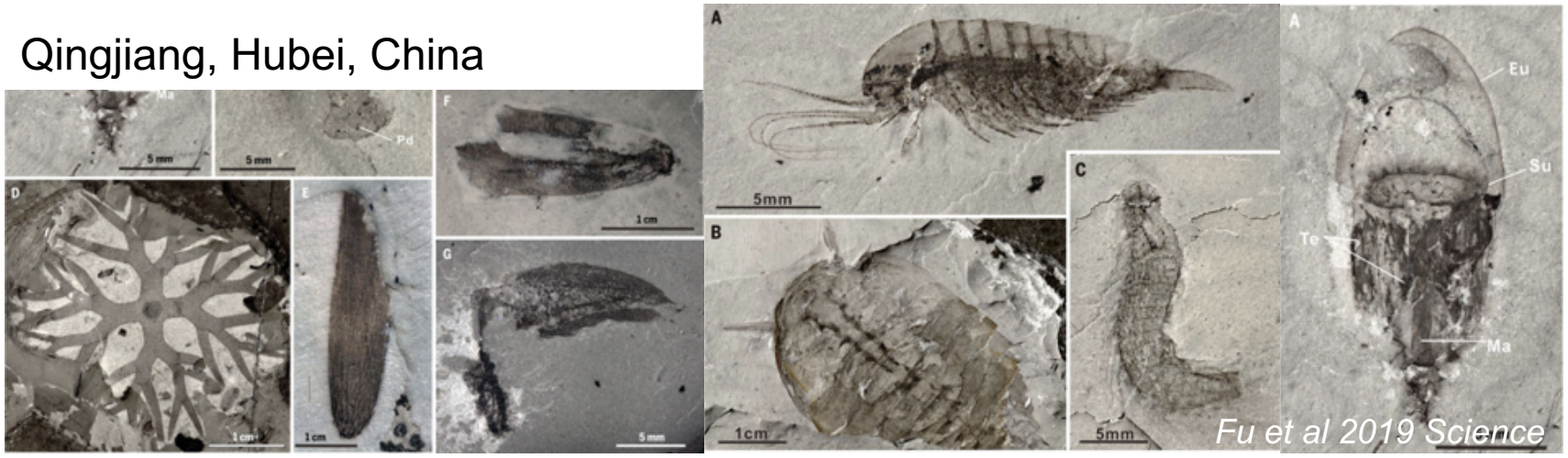
Weird looking things with lots of eyes, long proboscis

Chinese Burgess Shale Type

Chengjiang, Yunnan, China



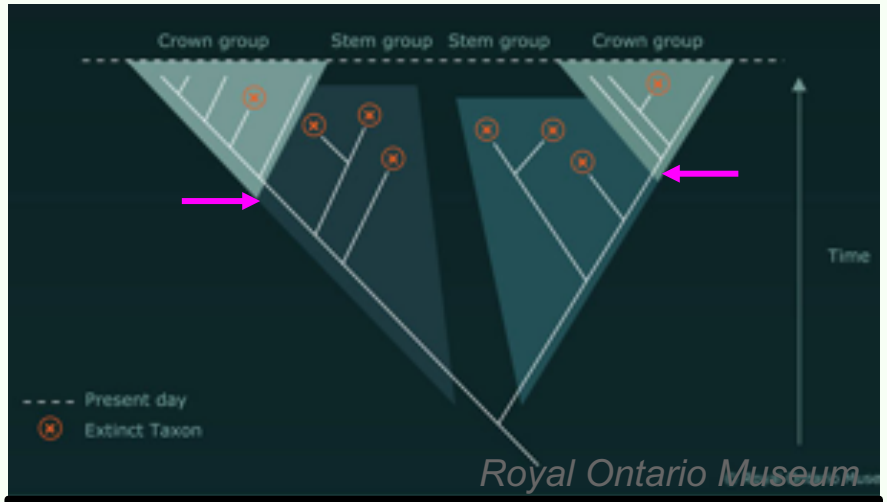
Qingjiang, Hubei, China



The Burgess Shale

Stem-groups

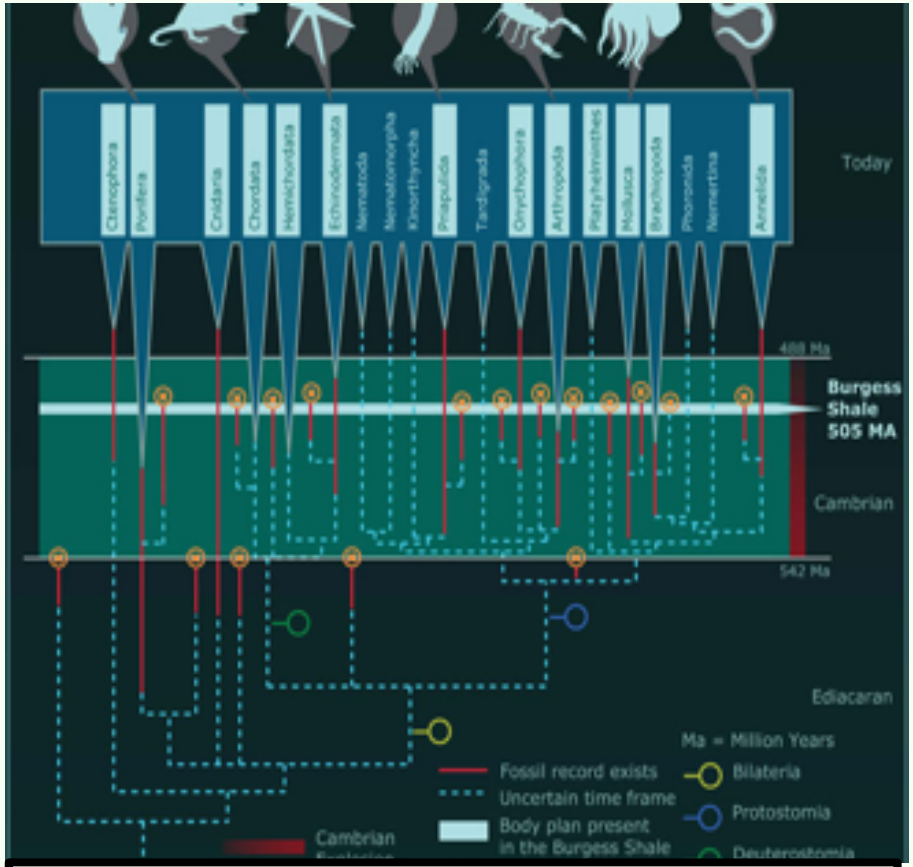
A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



The **crown group** is all living members of a clade, its common ancestor, and all descendants

The **stem group** are extinct organisms outside that group, on the lineage to that most recent common ancestor

Can be applied at all taxonomic levels

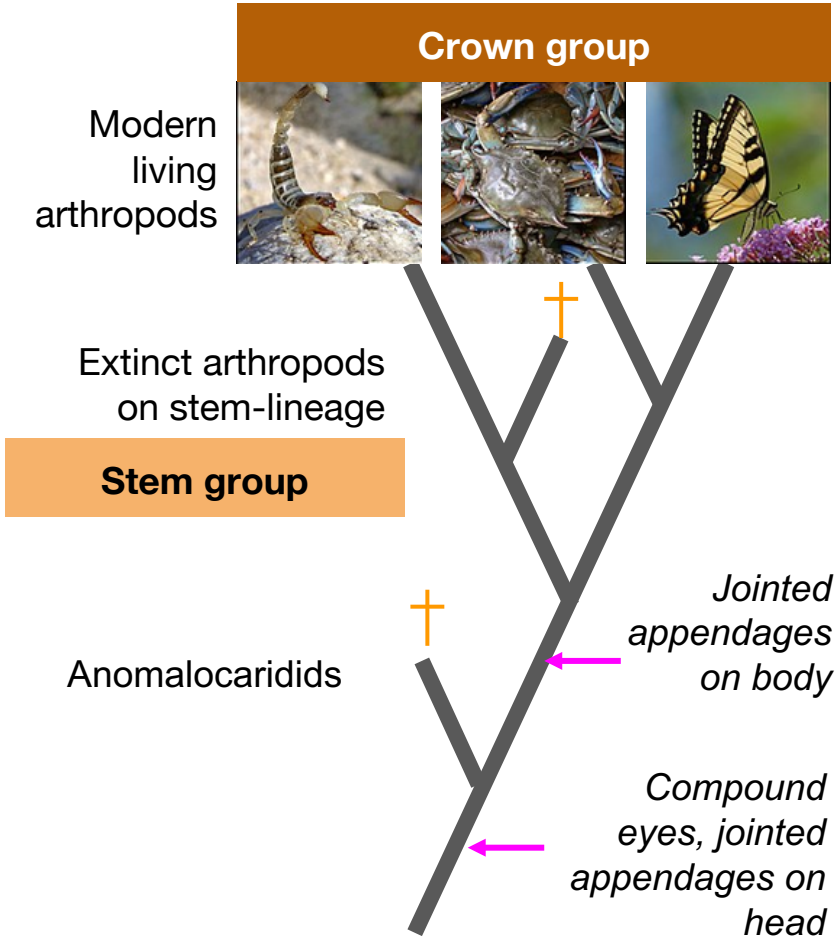


Many of the the Cambrian Burgess Shale fossil taxa are interpreted as **stem-groups of modern phyla**

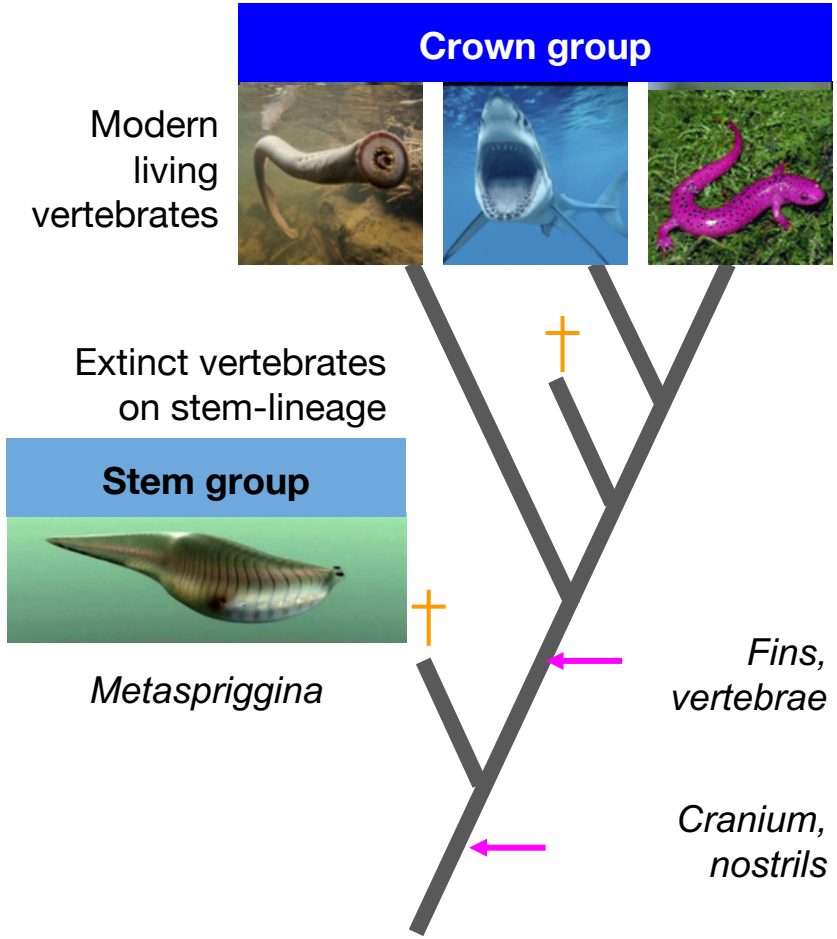
Stem-groups Examples

A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species

Arthropoda



Vertebrata



Origins of Animals
3.5 The Burgess Shale

Intended Learning Objective: Describe the Burgess Shale type fossil fauna, and interpretations of their biological affinity

Intended Learning Objective: Classify modern animals using the phylum concept, and extinct organisms using the stem-group concept

Robert Sansom, University of Manchester

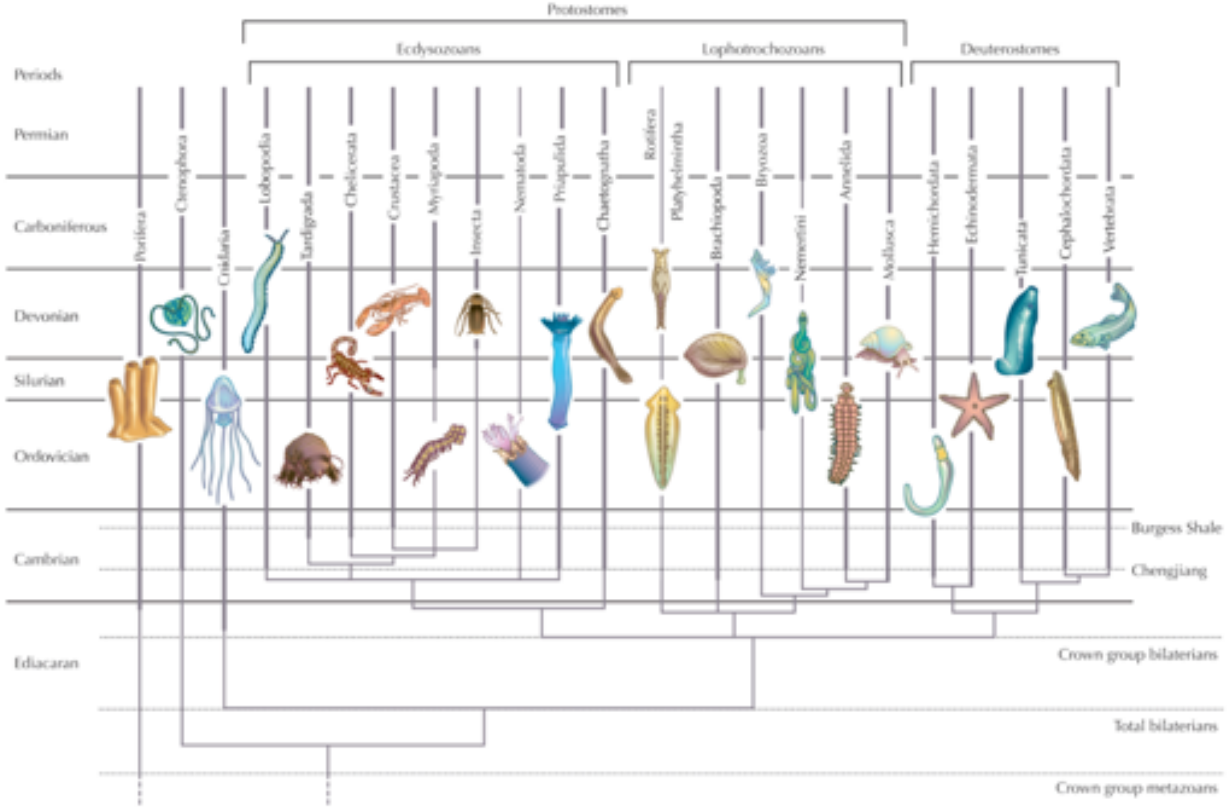
Part 3
The Origins of Animals

3.6 Cambrian Explosion

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Cambrian “Explosion”

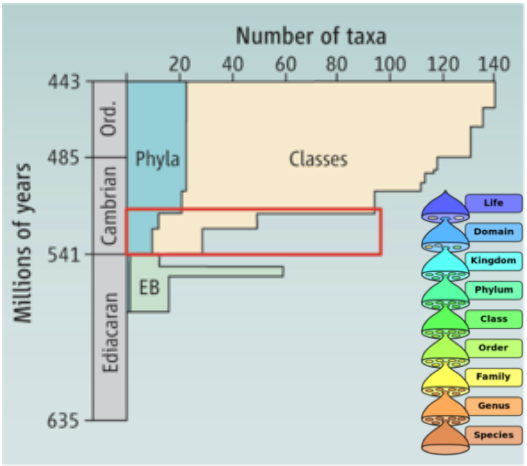
Phylogeny of animal phyla through time with known fossil ranges



Wills/Cold Spring Harbor

Most animal phyla first appear for the first time in a sudden burst of diversity: the **Cambrian Explosion**
Very few phyla have a later origin

Diversity of phyla and classes through time

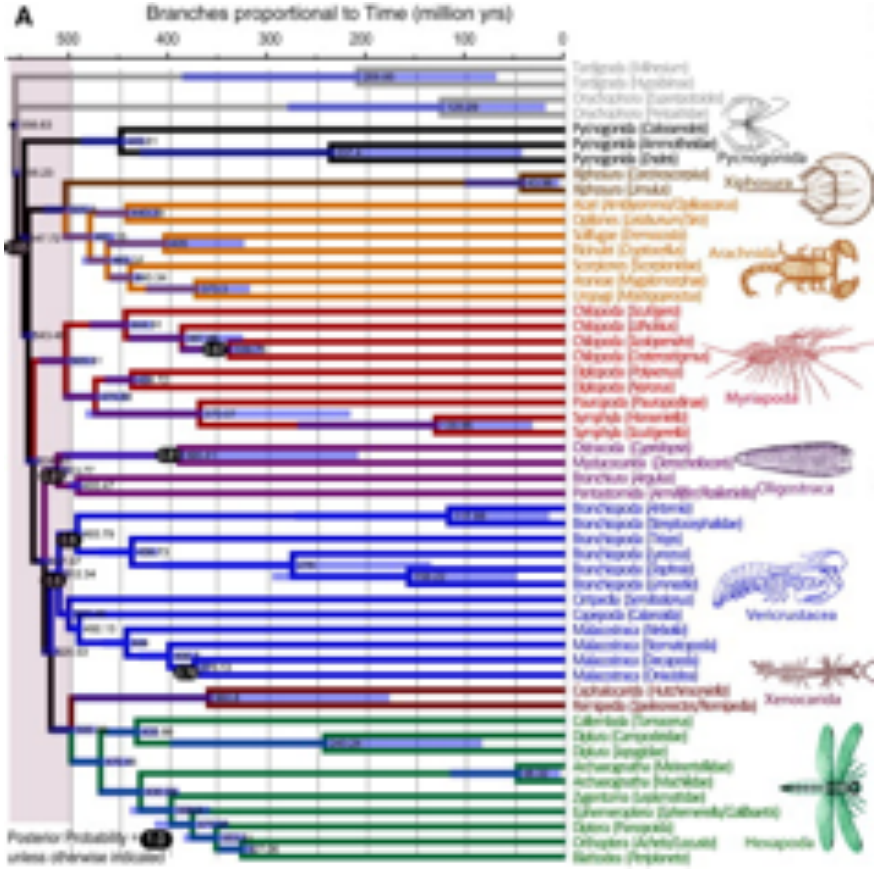


Smith 2013 Science

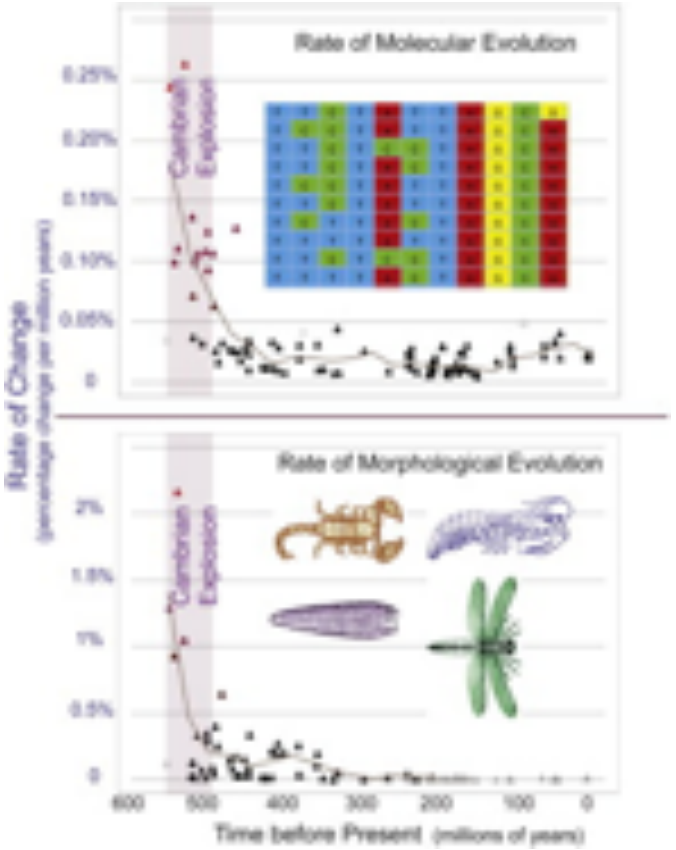
The diverse and disparate body plans of animals were therefore established in a very small period of time, at the beginning of the Cambrian (or earlier)

Rates of Evolution: A case study

An analysis of rates of arthropod evolution in the Cambrian and post-Cambrian



A massive phylogeny of arthropods using lots of molecular and morphological data



Evolution was about 5x faster than normal during the Cambrian

Lee et al 2013 Current Biology

Possible Cambrian Explosion Causes: Biological

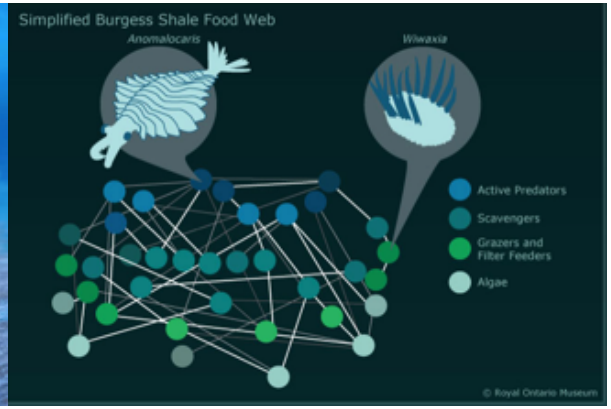
More complex ecological interactions: Predator-prey “arms race”

The crab is the natural predator of the snail

Natural selection favours snails with thicker shells and spines

Crabs evolve more powerful claws to pierce snails thicker shells

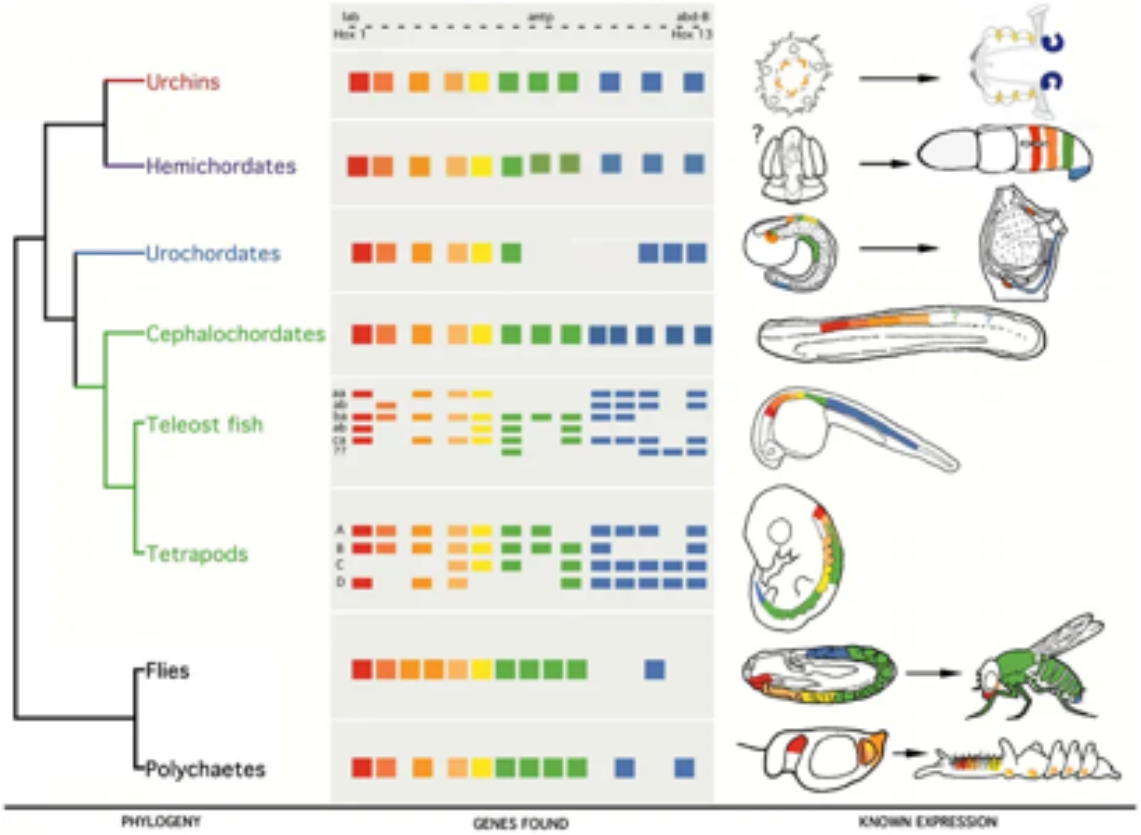
In response, natural selection favours snails with even thicker shells and spines



Cambrian ecosystems were more complex than before: predators and prey with complex visual and movement systems

Possible Cambrian Explosion Causes: Biological

Developmental "Explosion" of novelty: Newer body plans were easier to develop earlier on without constraints



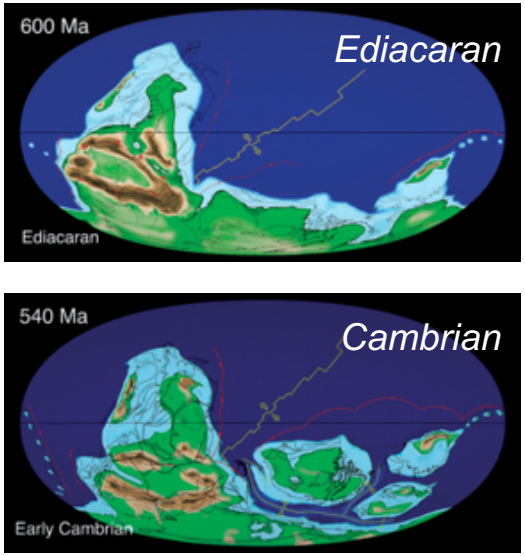
Animals all share a common fundamental mechanism by which they develop and grow: the pattern in which **Hox genes** are expressed in the developing embryo

Hox genes specify, for example, the position of head and appendages

Swalla 2006 Heredity

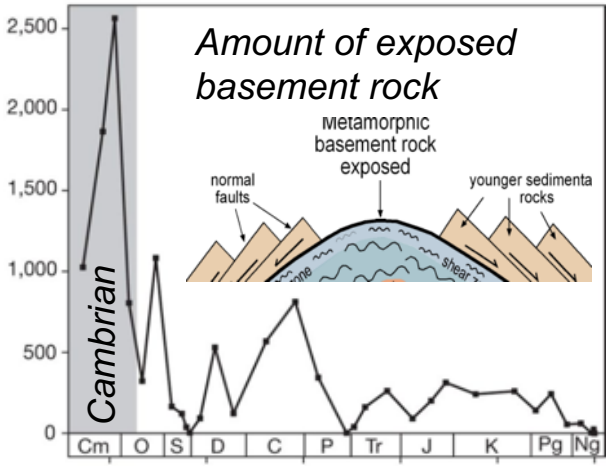
Earlier on, when there is no competition and no constraints, lots of different body plans might rapidly originate using same developmental mechanisms

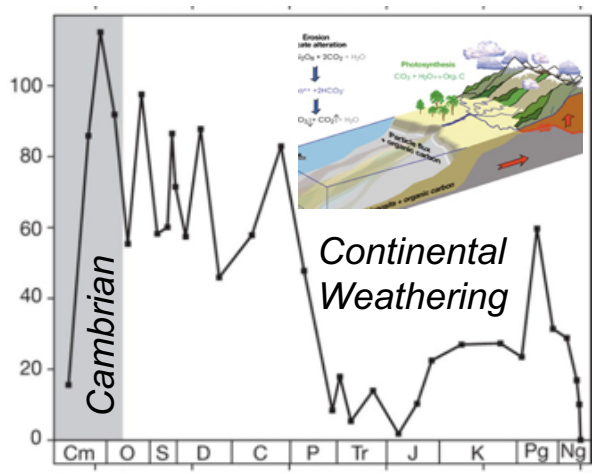
Possible Cambrian Explosion Causes: Geological



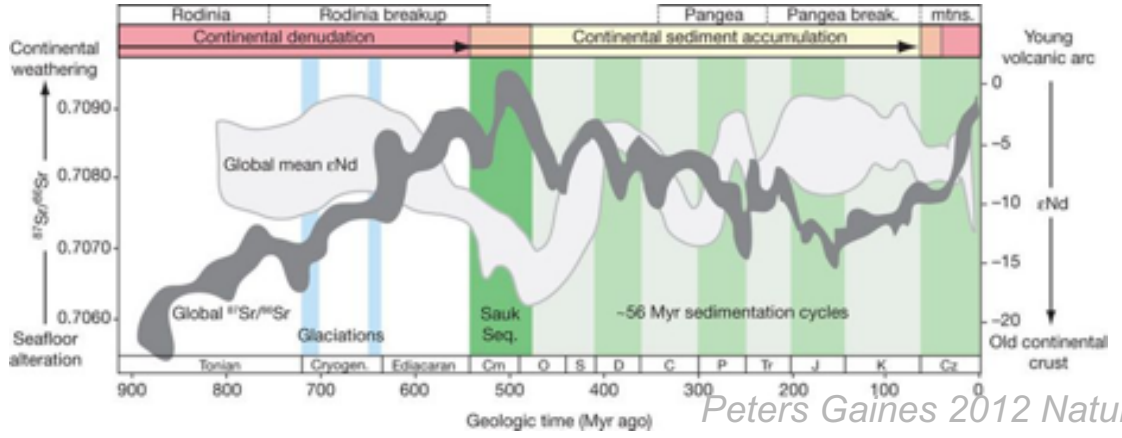
The arrangement of continents changes at beginning of Cambrian

Breakup of Rodina supercontinent and lots of exposed shelf: habitat for animals



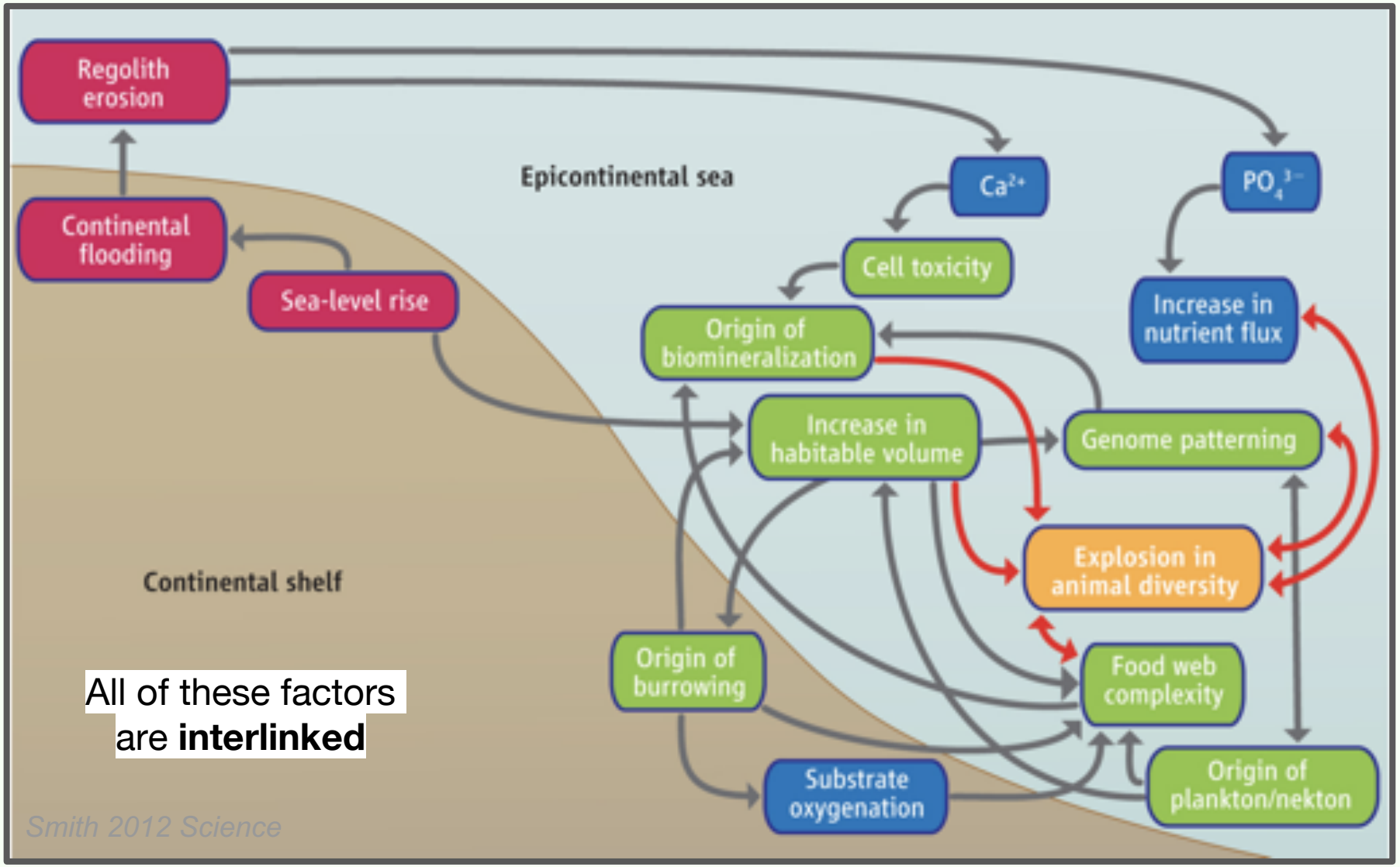


Large amount of exposed continental rock leads to much larger amount of weathering and carbonate input into the oceans and increasing ocean alkalinity



Peters Gaines 2012 Nature

Possible Cambrian Explosion Causes



Cambrian summary

- Sudden appearance of complex animal life in the Cambrian as demonstrated by the Burgess Shale
- Many of the fossil organisms belong to the stem groups of modern animal phyla
- Causes of this ‘explosion’ is complex interplay of biological and geological factors

Origins of Animals
3.6 Cambrian Explosion

Intended Learning Objective: Appraise evidence of the Cambrian 'explosion' and its possible biotic and abiotic causes

Question: How can we test hypotheses for causes of the Cambrian explosion?

Robert Sansom, University of Manchester